

COMP 4423: INTRODUCTION

Easy Computer Vision

Xiaoyong Wei (魏驍勇)
x1wei@polyu.edu.hk

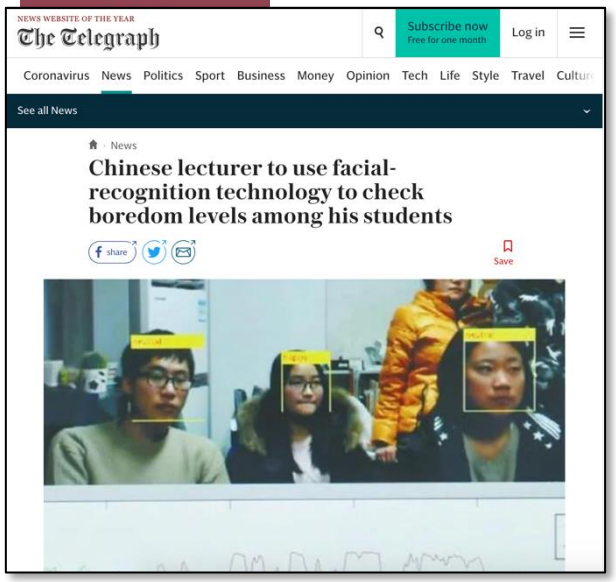
The Hong Kong Polytechnic University
香港理工大學
Opening Minds • Shaping the Future
啟迪思維 • 成就未來

The Lecturer

THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學

- Professor (Practice), Dept. of Comp., PolyU, HK
- Visiting Professor, Tampere University, Finland
- Adjunct Professor, Peng Cheng Lab
- Professor, Department Head, Sichuan University
- Postdoctoral Fellow, UC Berkeley
- PhD, City University of Hong Kong
- Founding Member of VIREO
- Senior Member of IEEE

PolySmart Group



The Telegraph Sep. 13, 2016



Digital Trends Sep. 13, 2016



XinHua News May 25, 2018



CCTV Jan. 13, 2013



The Paper May 27, 2018



QUARTZ Sep. 13, 2016



China Youth Daily Jun. 13, 2018



China News May 26, 2018



Shenzhen TV Mar. 2, 2020



China Youth Net Jun. 04, 2013



“Pineapple Science Prize” 2013
Awarded by Nobel Prize laureate



Xinhua News Oct. 25, 2012

Office Hours

THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學

- Every Tuesday 16:00 -18:00 at PQ829
- Appointments via x1wei@polyu.edu.hk or by phone 2766-7258

Languages

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- English (Professional)
- Mandarin (Native)
- Cantonese (Pretend to be native)

Teaching Assistants

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- Wengyu Zhang, 25015359r@connect.polyu.hk



What is **Computer Vision**?



I have no idea …


Freedom of Speech





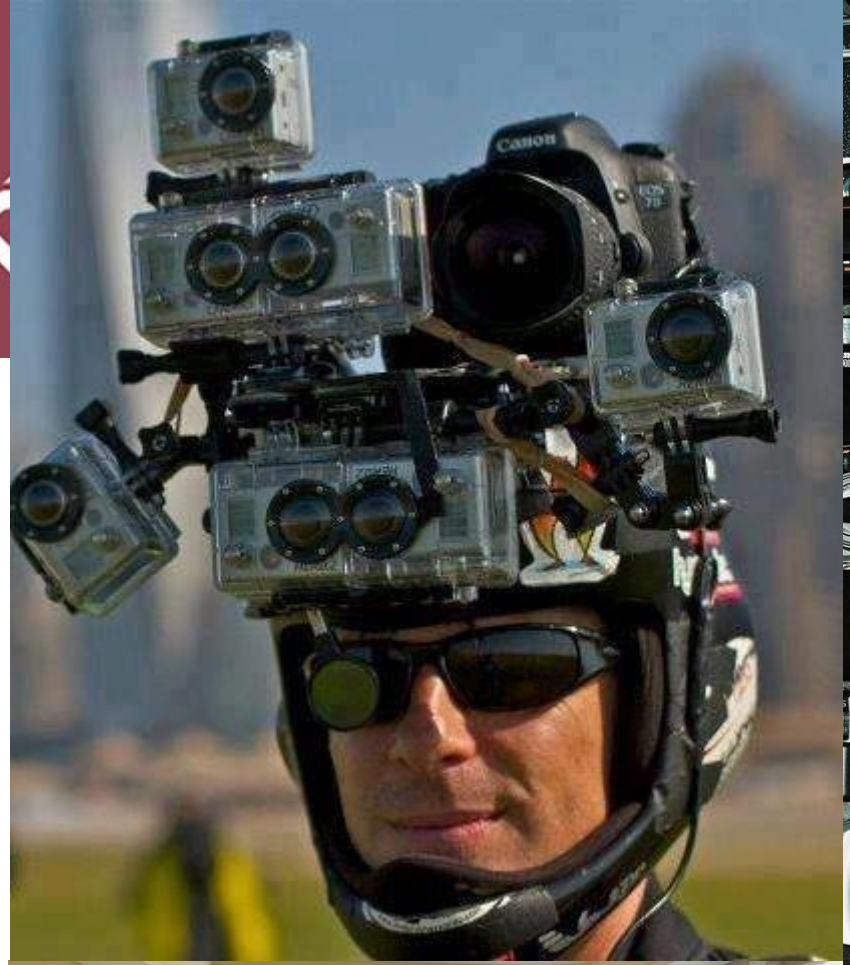











Canon

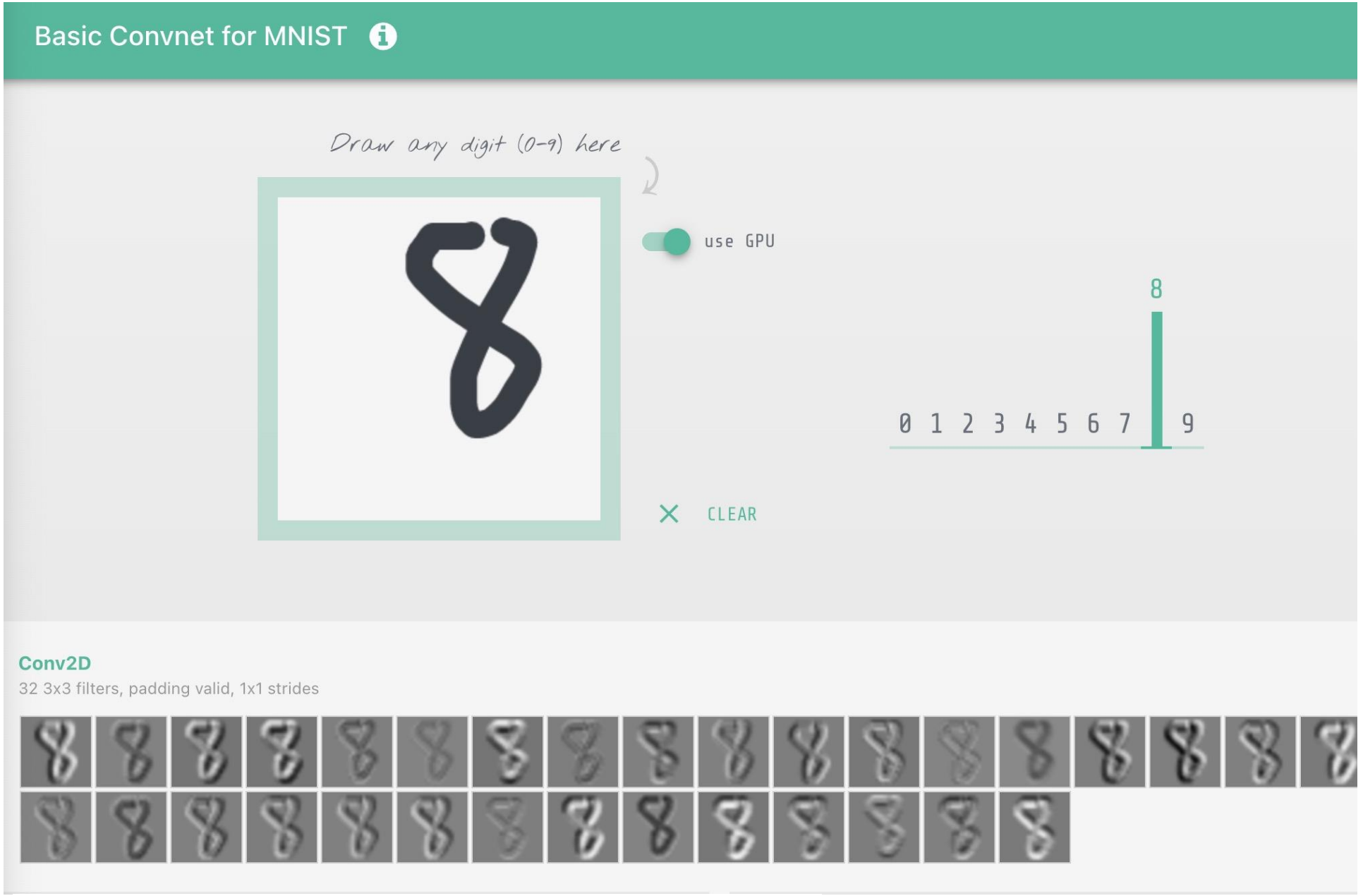



Kodak
VR35
Konica
Canon
OLYMPUS
M7
CONTAX
Mamiya
CHINON
PENTAX
MINOLTA
NIKONOS


Edge Detection
Convolution
Corners
SIFT
HOG
Image Filtering
Image Filtering
Receptive Field
Pose Estimation
Corners
Pooling
Bounding Box
Anchor Box
Non-Maximum Suppression
Image Filtering
Non-Maximum Suppression
U-Net
ResNet
VGG
Faster R-CNN
Semantic Segmentation
Panoptic Segmentation
U-Net
RetinaNet
Contrastive Learning
CLIP
YOLO
Mask R-CNN
SSD
RetinaNet
Visual SLAM
Computer Vision
RetinaNet
DETR
Diffusion Models
Opereic Flow
SAM
Adversarial Examples
Object Detection
Image Segmentation
DETR
ImageNet
GAN
Transformer
CNN
Transformer
Feature Extraction
VGG
RetinaNet
Zero-Shot Learning
Self-Supervised Learning
Multimodal Learning
Domain Adaptation
Contrastive Learning
Vision Transformer
Style Transfer
Image Captioning
Visual Question Answering
Multimodal
Style Transfer
Epipolar Geometry
Calibration
Dataset Bias
OCR
Super-Resolution
Denoising
Anomaly Detection
Dataset Bias
Saliency



But let’s play with it!



https://transcranial.github.io/keras-js/#/mnist-cnn

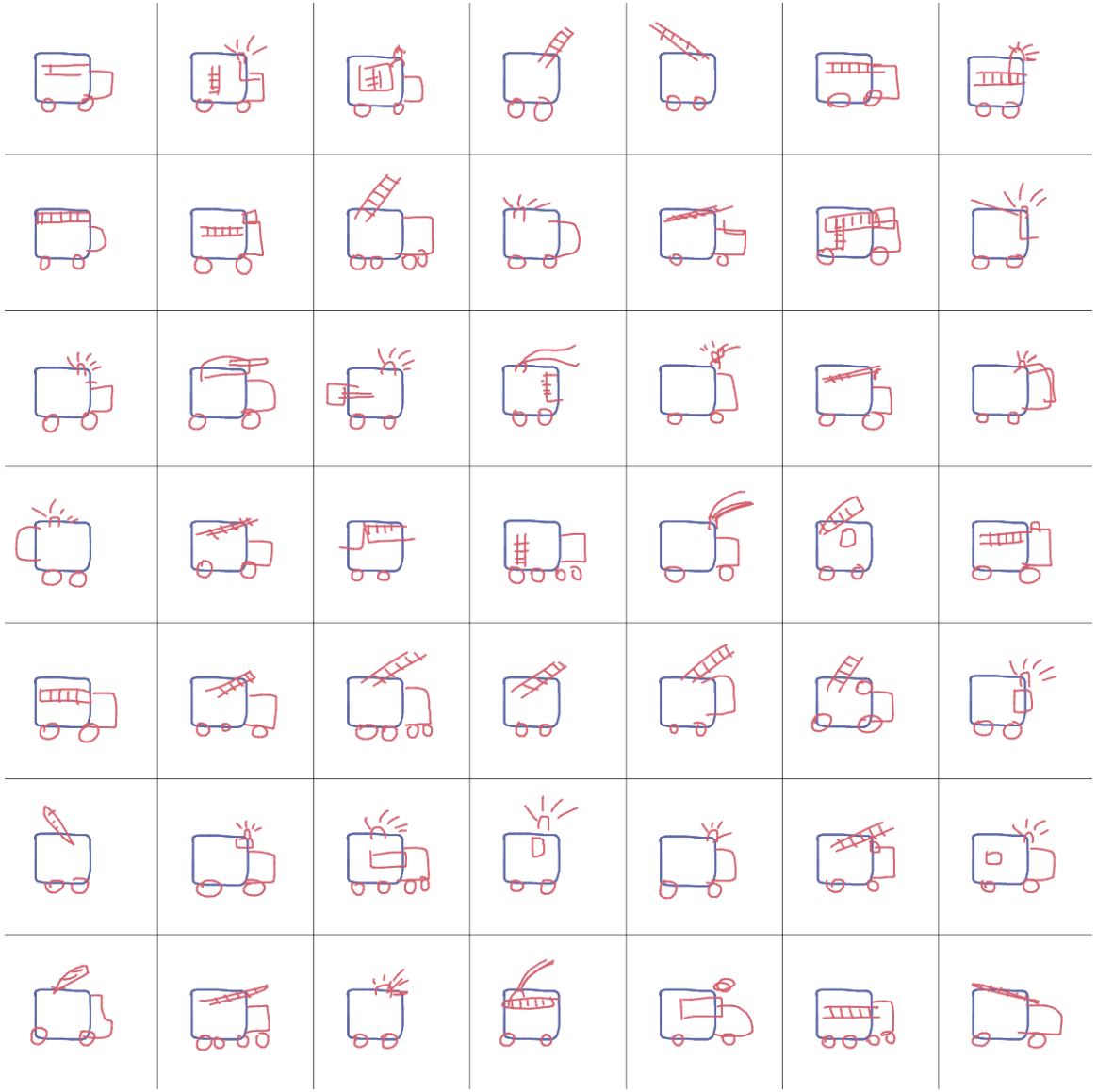


Can a neural network learn to recognize doodling?

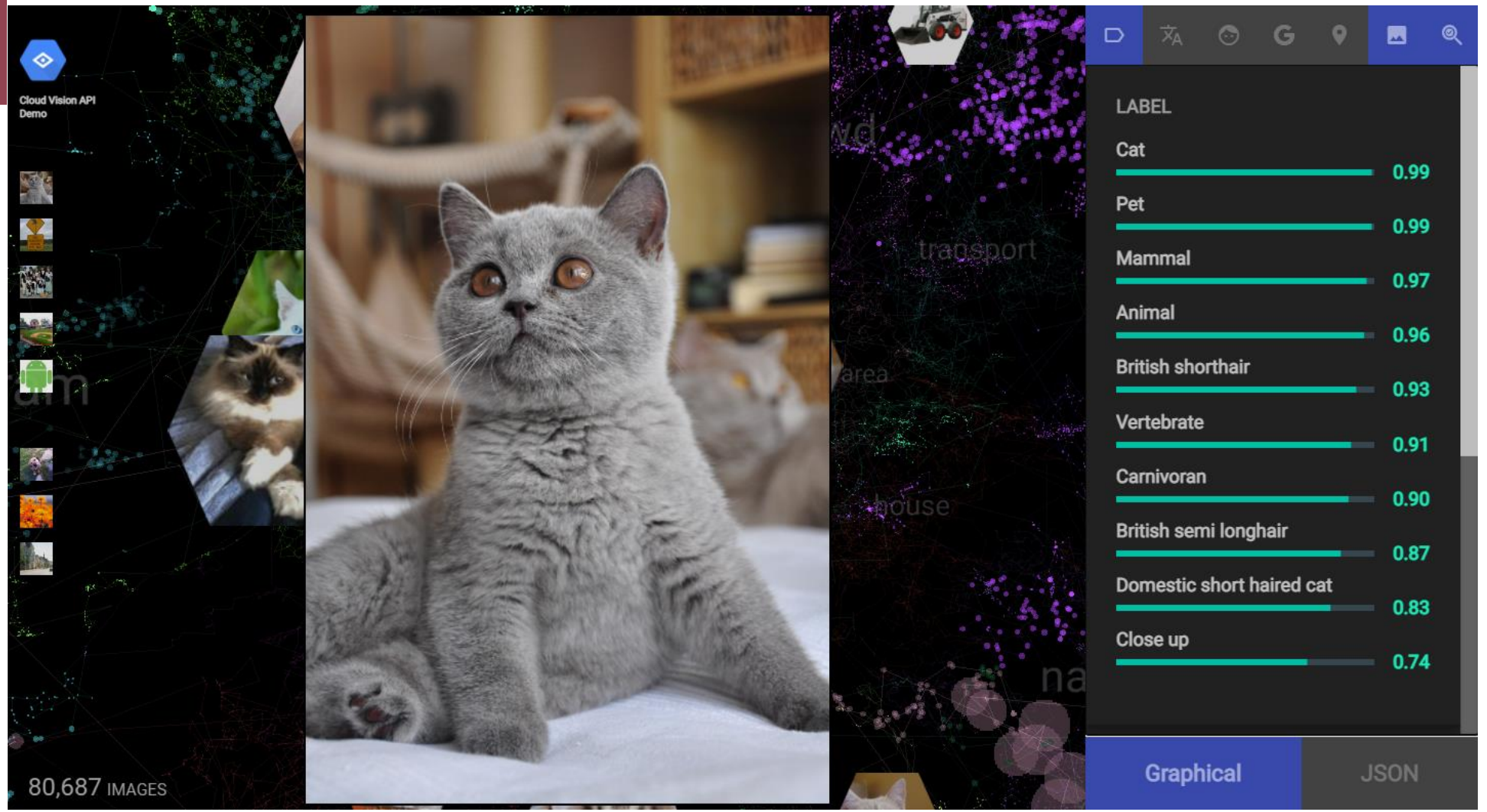
Help teach it by adding your drawings to the world's largest doodling data set, shared publicly to help with machine learning research.

Let's Draw!

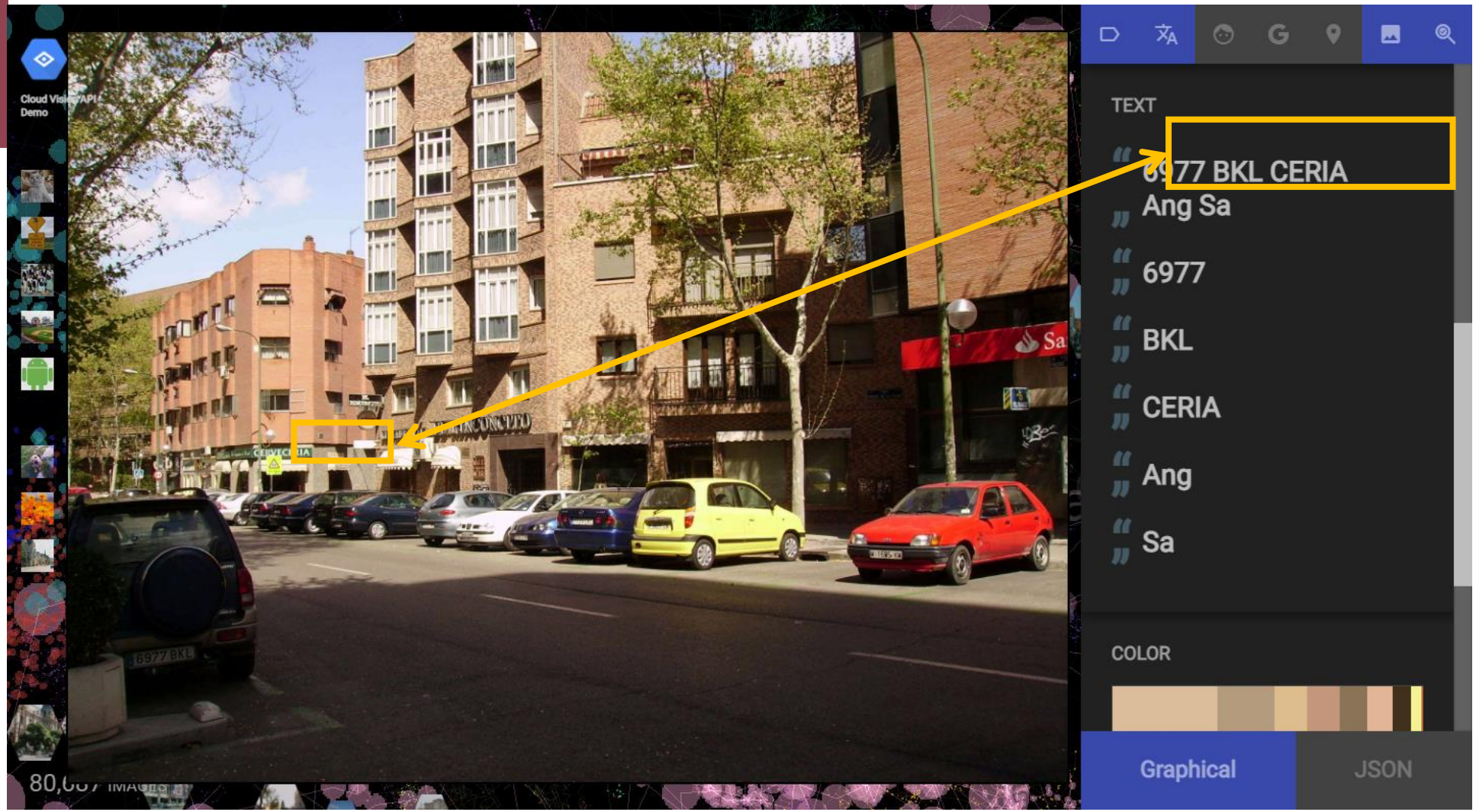
https://quickdraw.withgoogle.com/#



https://magenta.tensorflow.org/sketch-rnn-demo



http://vision-explorer.reactive.ai/#/galaxy?_k=8d4jqd



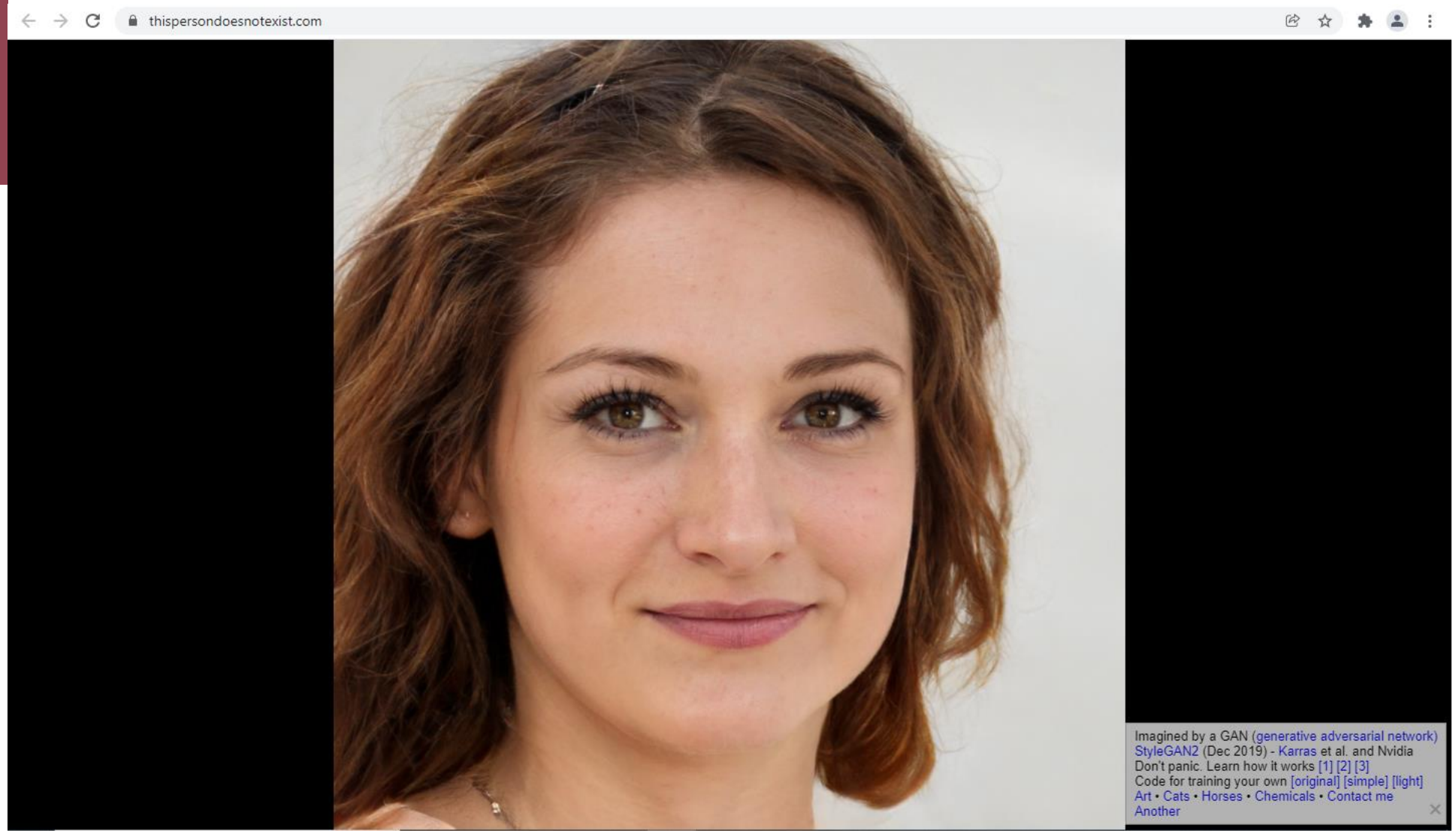
http://vision-explorer.reactive.ai/#/galaxy?_k=8d4jqd











https://thispersondoesnotexist.com/







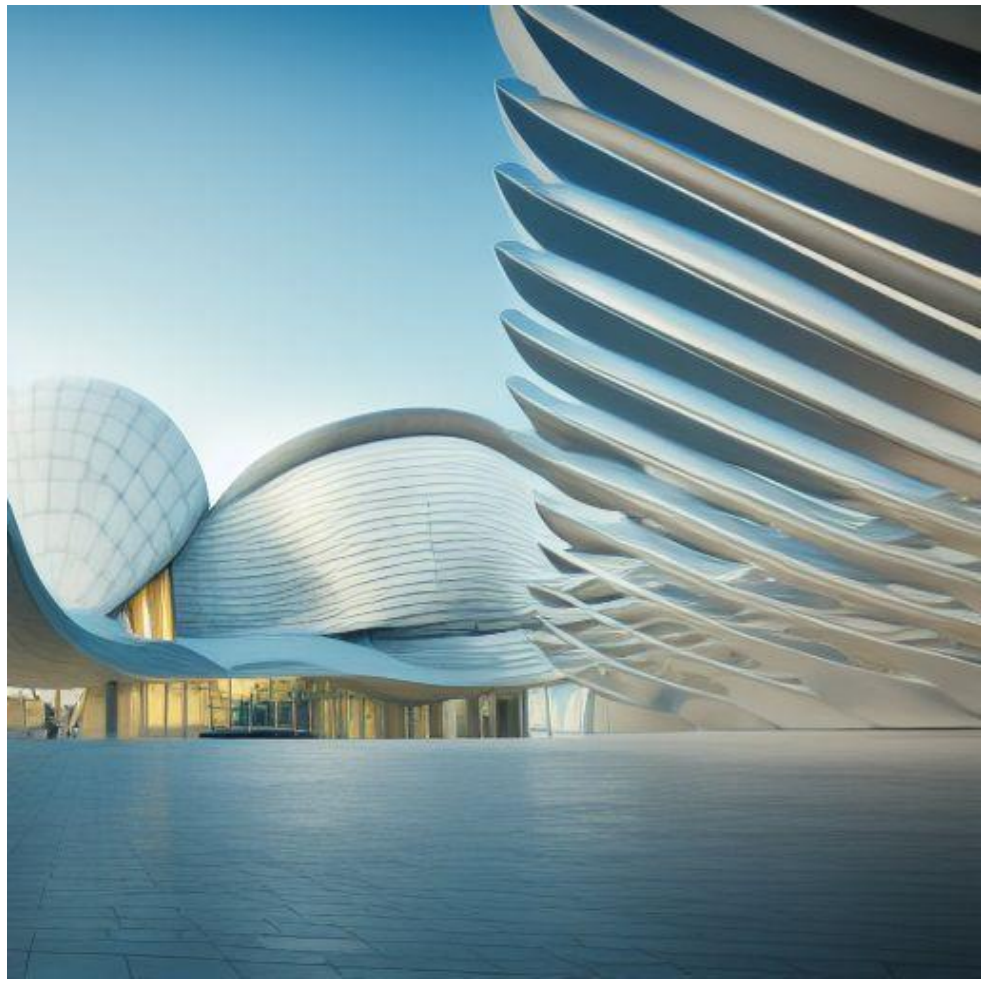














https://stablediffusionweb.com/

Try It

interior design, frank lloyd wright house cave with forest canopy, dark wood, streaks of light, light fog, living room :: bubbletech –test –ar 9:16

cyberpunk techwear streetwear look and clothes, we can see them from feet to head, highly detailed and intricate, golden ratio, beautiful bright colors, hypermaximalist, futuristic, cyberpunk setting, luxury, elite, cinematic, techwear fashion, Errolson Hugh, Sacai, Nike ACG, Yohji Yamamoto, Y3, ACRNYM, matte painting –w 2176 –h 3840 –iw 1

woman, warrior, detailed illustration, digital art, overdetailed art, concept art, full character, character concept, long hair, full body shot, highly saturated colors, fantasy character, detailed illustration, hd, 4k, digital art, overdetailed art, concept art, Dan Mumford, Krzysztof Maziarz, trending on artstation

a full page design of spaceship engine, black and bronze paper, intricate, highly detailed, epic, infographic, marginalia –ar 9:16

https://stablediffusionweb.com/

AI Art at Christie's Sells for $432,500



"Edmond de Belamy, from La Famille de Belamy," by the French art collective Obvious, was sold on Thursday at Christie's New York. Christie's

Last Friday, a portrait produced by artificial intelligence was hanging at Christie's New York opposite an Andy Warhol print and beside a bronze work by Roy Lichtenstein. On Thursday, it sold for well over double the price realized by both those pieces combined.

"Edmond de Belamy, from La Famille de Belamy" sold for $432,500 including fees, over 40 times Christie's initial estimate of $7,000-$10,000. The buyer was an anonymous phone bidder.

The bidding late this morning lasted just under seven minutes, during which the buyer competed against an online bidder in France, two other phone bidders and one person in the room in New York. When the hammer came down, the bids had reached $350,000, the final price before fees.

The distorted portrait, by the French art collective Obvious, was marketed by Christie's as the first portrait generated by an algorithm to come up for auction.

https://www.nytimes.com/2018/10/25/arts/design/ai-art-sold-christies.html



Applications

Applications

https://www.tesla.com/autopilot
https://en.wikipedia.org/wiki/Self-driving_car

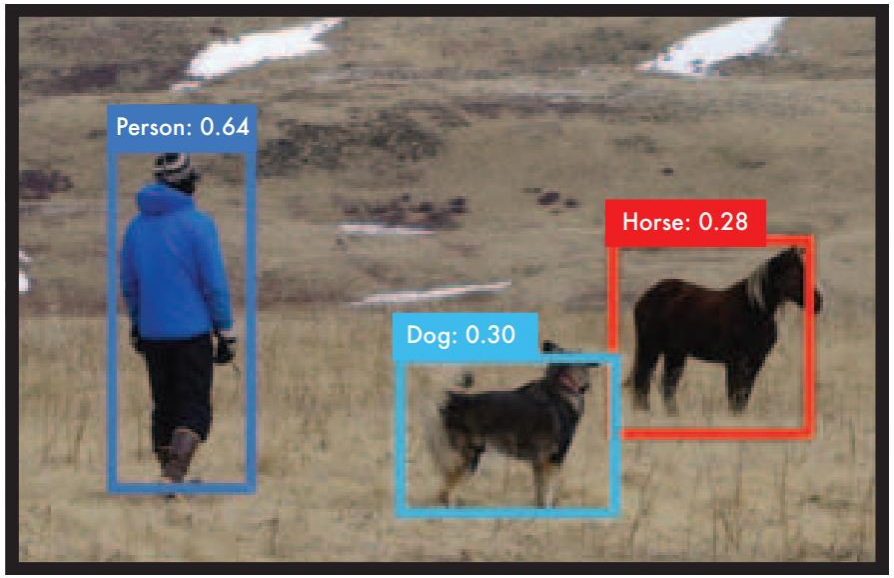



Vehcle (9.3m)
ID:9
Vehcle (16.7m)
ID:40
Vehcle (24.6m)
ID:15
Vehcle (23.3m)
ID:39
Vehcle (8.4m)
ID:0
Lane: Left
V:11.7mph
Stationary
Lane: Imm-Left
V:10.2mph
Stationary
Lane: Imm-Left
V:11.0mph
Stationary
Lane: Imm-Left
V: 7.4mph
Stopped
Lane: My
V:-0.9mph
Moving

Applications

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The task of **Object Detection** is to find out all the objects of interest in the image and determine their categories and positions. Object detection is always the most challenging problem in the field of computer vision because of the different appearance, shape and posture of objects and the interference of illumination and occlusion.



Redmon J , Divvala S , Girshick R , et al. You Only Look Once: Unified, Real-Time Object Detection[C]// Computer Vision & Pattern Recognition. IEEE, 2016.

Applications

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Semantic Segmentation, or image segmentation, is the task of clustering parts of an image together which belong to the same object.

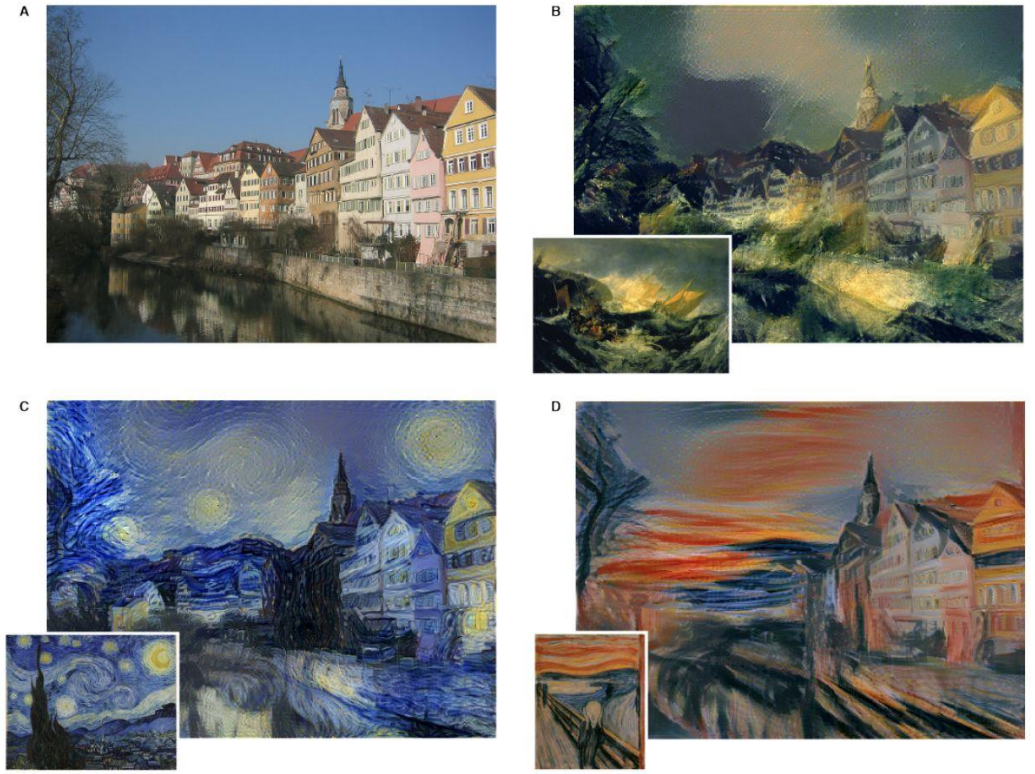


M. Everingham, L. Van Gool, C. K. Williams, J. Winn, and A. Zisserman. The pascal visual object classes (voc) challenge. IJCV, 88(2):303–338, 2010.

Applications

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Style Transfer refers to applying the style of one domain or several images to other domains or images. For example, abstract styles are applied to realistic images.



Gatys L A , Ecker A S , Bethge M . A Neural Algorithm of Artistic Style[J]. Journal of Vision, 2015.



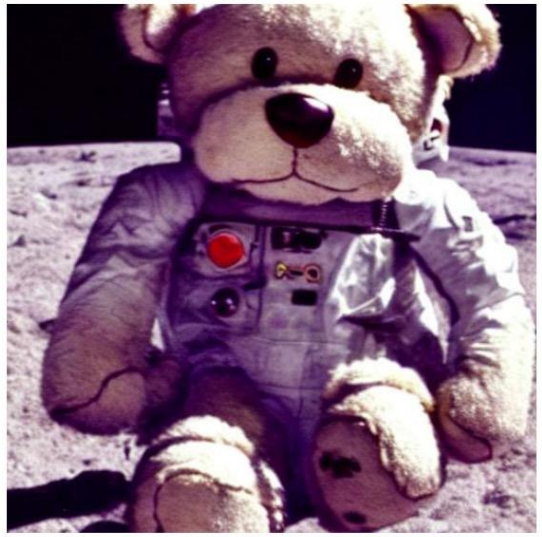
dog* under a hot air balloon



dog* playing ball on the beach



dog* and a balloon



teddybear* in a spacesuit



teddybear* playing the guitar



teddybear* in Venice



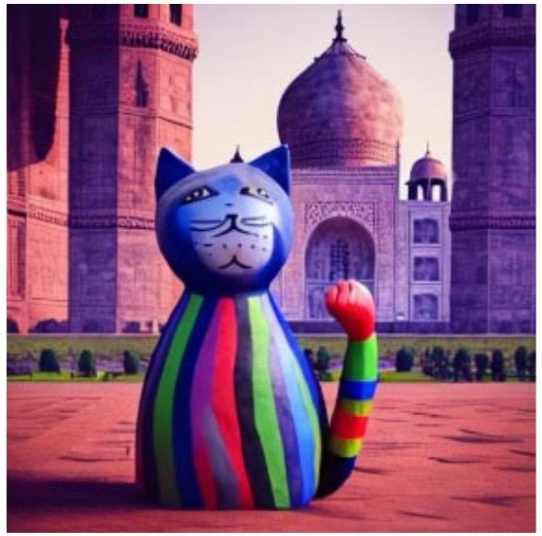
wooden pot* in the hands of an Egyptian pharaoh



cat* in a spaceship



wooden pot* next to the Sphinx



cat statue* in front of Taj Mahal



cat statue* in desert



dog* in couch

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Compositional Inversion for Stable Diffusion Models, The 38th Annual AAAI Conference on Artificial Intelligence (AAAI), 2024

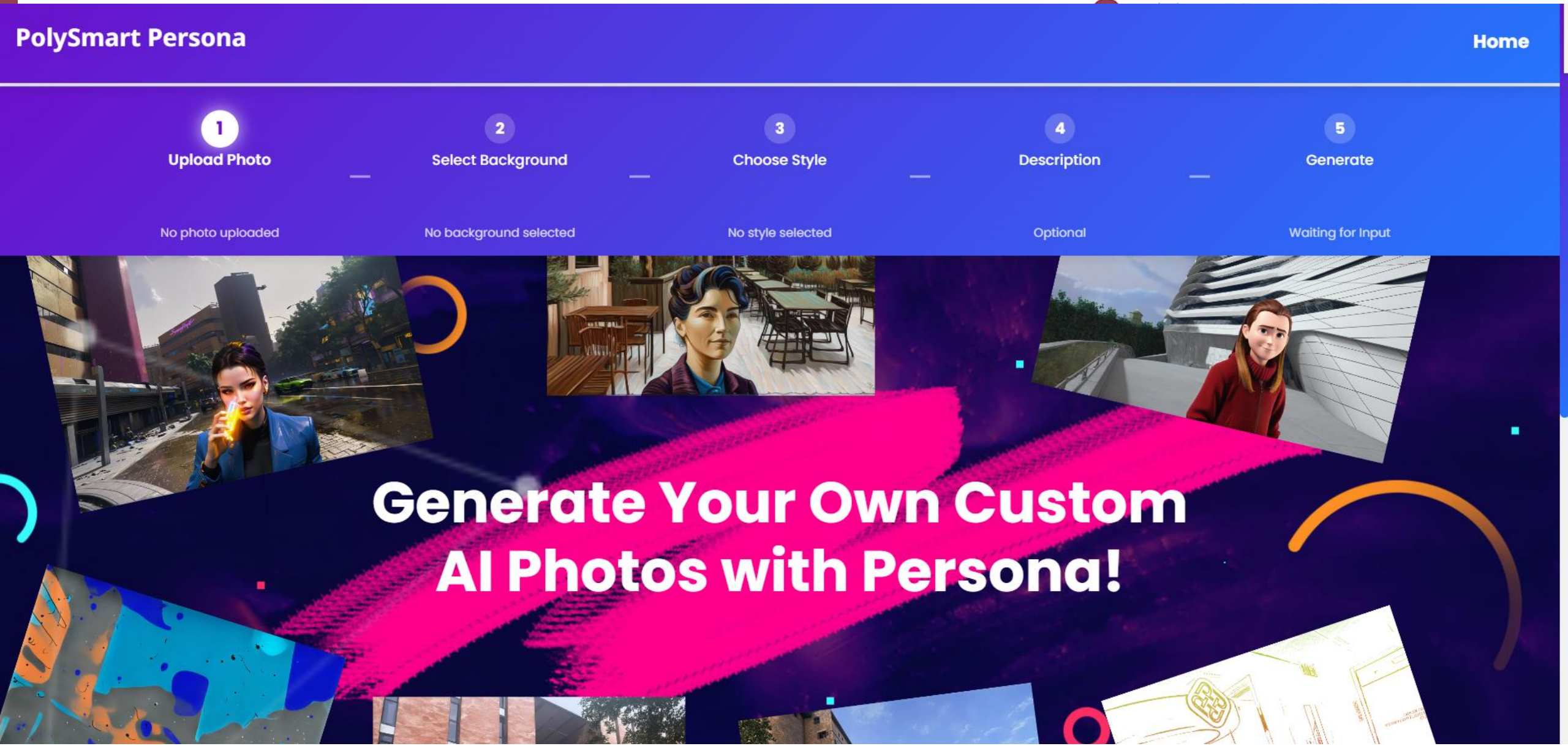


Generative active learning for image synthesis personalization, X Zhang, W Zhang, X Wei, J Wu, Z Zhang, Z Lei, Q Li
Proceedings of the 32nd ACM International Conference on Multimedia, 10669-10677



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Do NOT Distribute

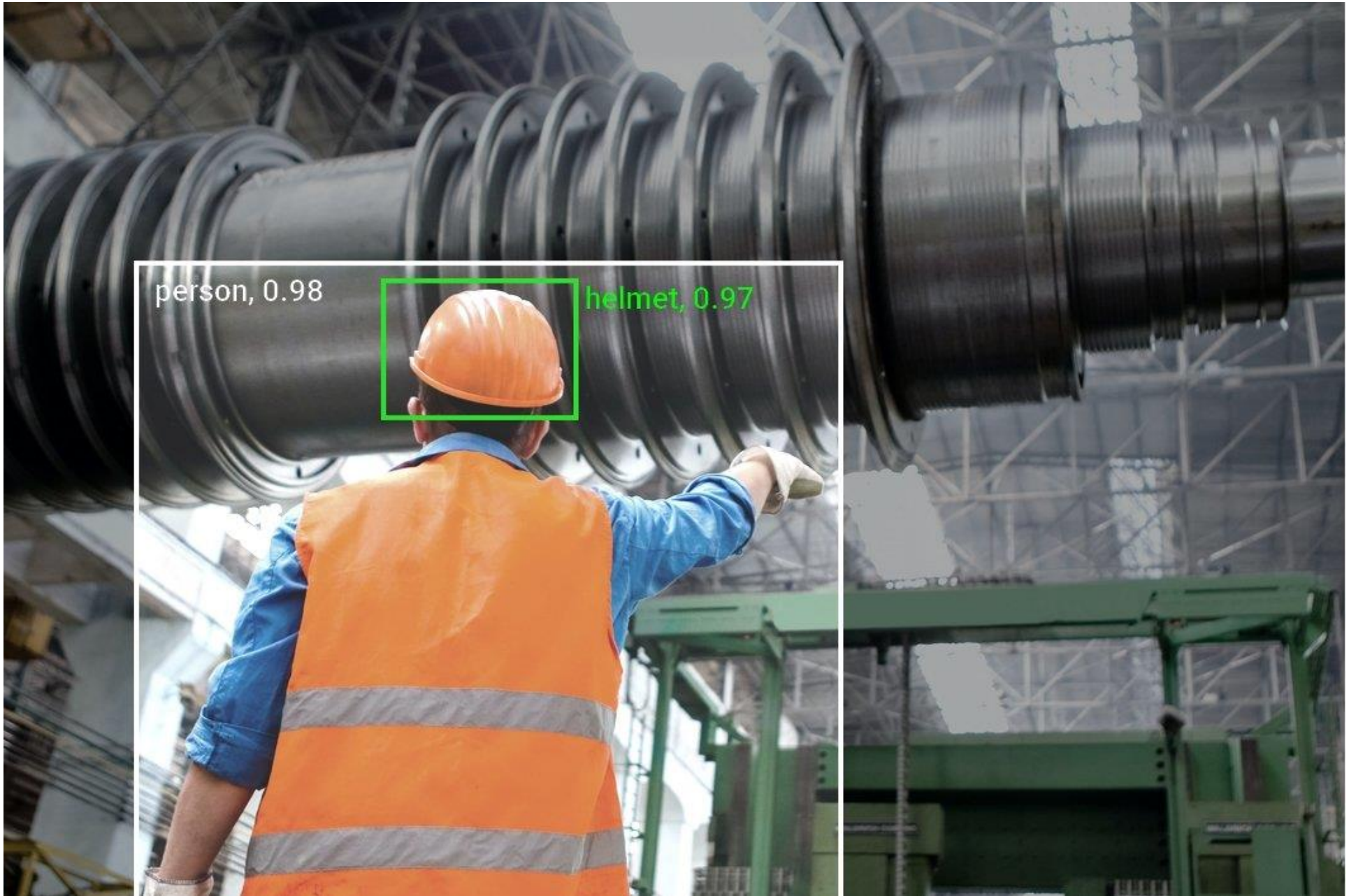

PolySmart Persona
Home
1
Upload Photo
No photo uploaded
2
Select Background
No background selected
3
Choose Style
No style selected
4
Description
Optional
5
Generate
Waiting for Input
Generate Your Own Custom
AI Photos with Persona!


Faculty of Computer and Mathematical Sciences
計算機及數學科學學院
理大校友日2026
POLYU HOMECOMING 2026
AI-ART
PolySmart Persona
在Poly的U，遇見Poly的你
Generate Your Own Custom
AI Photos with Persona!
Department of Computing
電子計算學系
POLYU ALUMNI


Faculty of Computer and Mathematical Science
計算機及數學科學學院
OMNICOLOR
SKECHERS
Enjoy the Comfort


Faculty of Computer and Mathematical Sciences
計算機及數學科學學院
OMNICOLOR

GEARBOX - 99%
TOOLS - 93%
TOOLS - 94%
FORKLIFT - 98%
ENGINE - 96%
FORKLIFT - 95%
PERSON - 99%
ENGINE - 97%
PERSON - 94%
PERSON - 98%
PERSON - 97%
TOOLS - 97%


person, 0.98
helmet, 0.97

X = 0.01129 inch; Y = 0.00824 inch; Diagonal = 0.01397 inch

ARTY
AVNET
DIGILENT
QC.OK



Driving



Surveillance



Medicine



Services



Smart Building



Entertainment



What is **Computer Vision**?



A brief history

1950s-1970s: Early Beginnings
1970s-1980s: Edge Detection and Feature Extraction, OCR
1980s-1990s: Object Recognition and Machine Learning
2000s: Emergence of Support Vector Machines and Viola-Jones Algorithm
Late 2000s-2010s: Deep Learning Revolution
2010s-Present: Advances in Deep Learning and Beyond

https://medium.com/@ambika199820/what-is-computer-vision-history-applications-challenges-13f5759b48a5

1950s-1970s: Early Beginnings



Russell's infant son

Work on computer image processing started much earlier, in **1957**, at the American National Institute for Standards and Technology, where a group of engineers, led by **Russell Kirsch**, succeeded in making the **first ever digital scan of an image**. The image they used was of Russell's infant son, which became so famous that the *Life* magazine included it in its article about 100 images that changed the world. The original image is being kept in the Portland Art Museum.

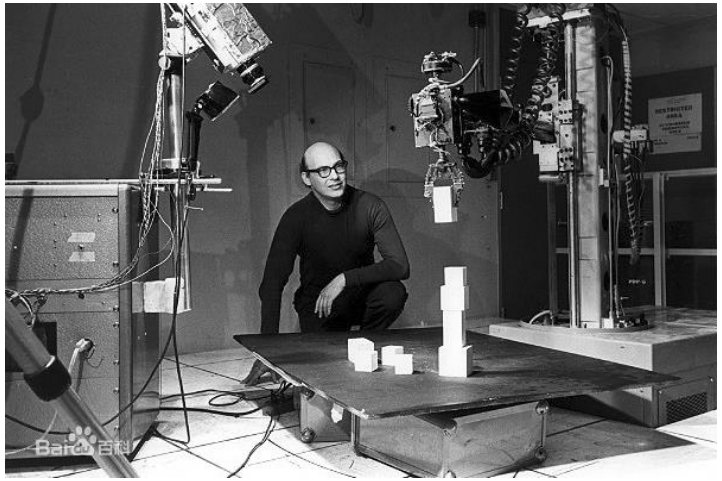
https://www.megatrend.com/en/the-history-of-computer-vision-and-the-evolution-of-autonomous-vehicles/

1950s-1970s: Early Beginnings

It was believed by some of the early pioneers of artificial intelligence and robotics (at places like MIT, Stanford, and CMU) that solving the "visual input" problem would be an easy step along the path to solving more difficult problems such as higher level reasoning and planning.

In 1966, Marvin Minsky at MIT asked his undergraduate student Gerald Jay Sussman to "spend the summer linking a camera to a computer and getting the computer to describe what it saw".

Marvin Minsky proposed that framework was the basis of visual perception. For example, here is a framework for cars as follows:

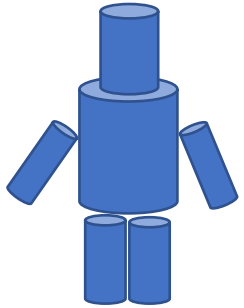


Marvin Minsky

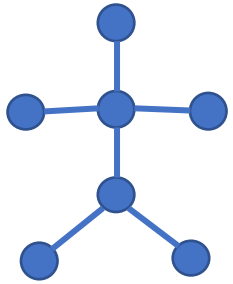
Name: Car
Super-class: Vehicle
Sub-class: Van, Jeep
Number of wheels: Unknown
Length: Unknown

A brief history – 1970s-1980s

How can we recognize and represent objects?



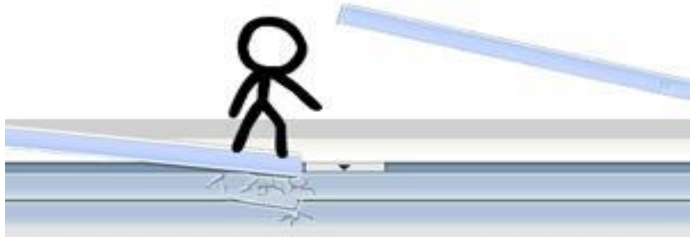
generalized cylinder
Brooks & Binford, 1979



pictorial structure
Fischler & Elschlager, 1973



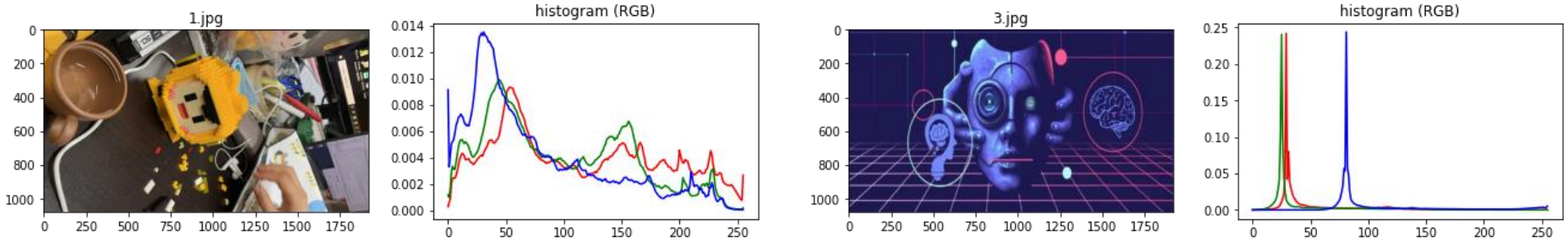
Minecraft



Matchstick Man

1970s-1980s: Edge Detection and Feature Extraction





1970s-1980s: OCR



1980s-1990s: Object Recognition and Machine Learning

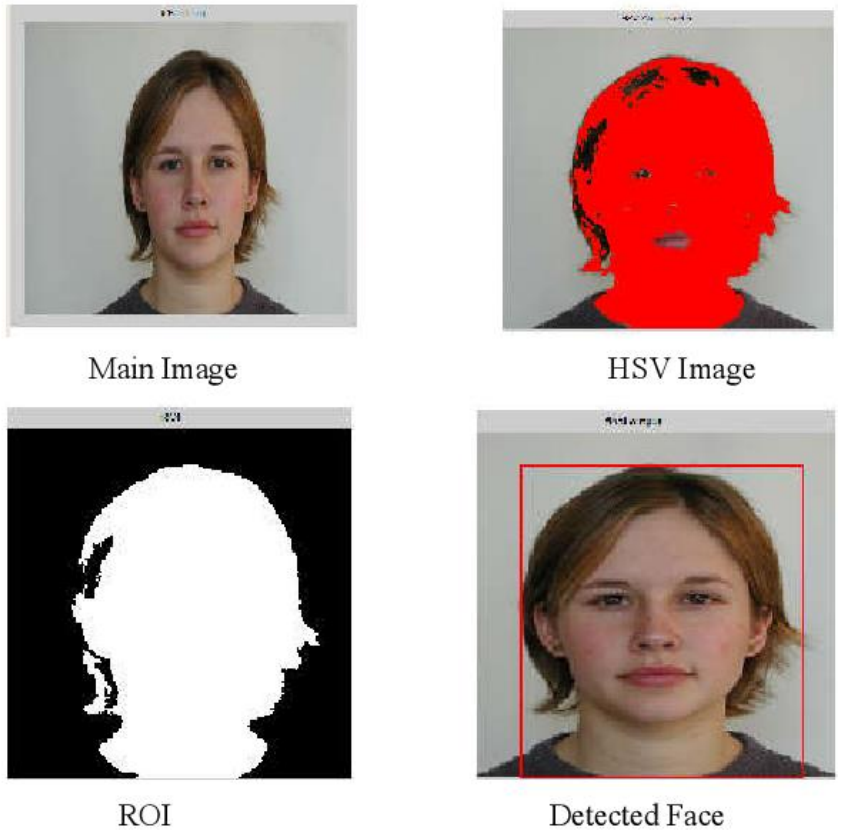
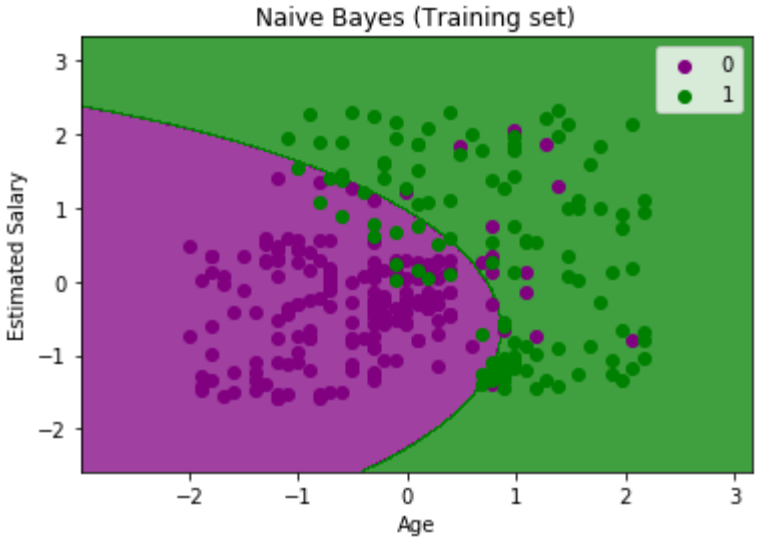
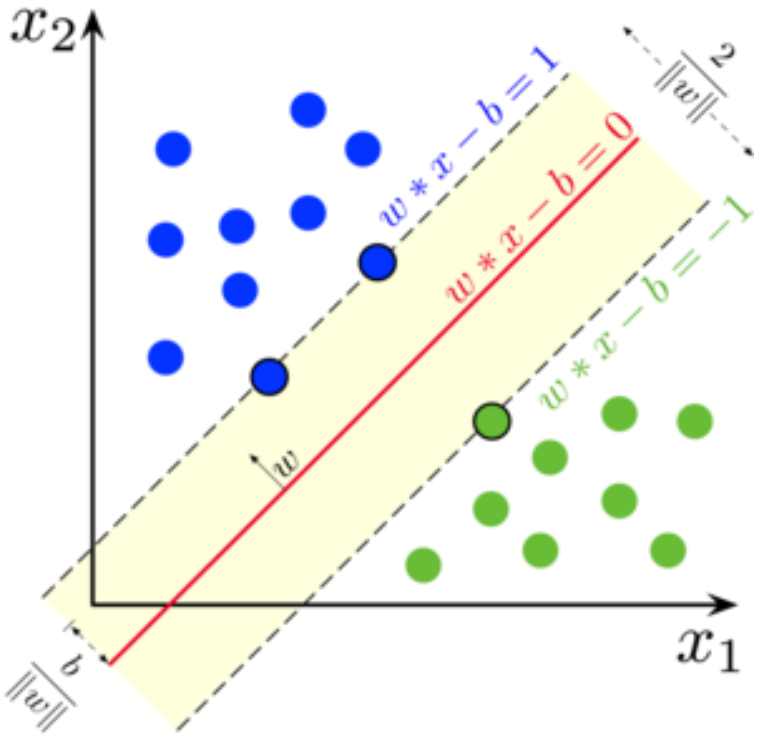
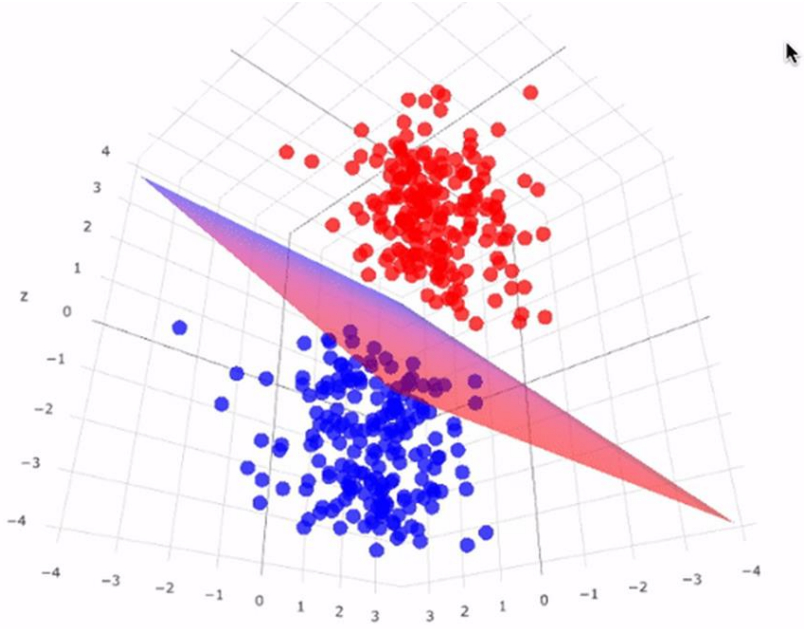


Fig 1 Skin Colour Based Face Detection

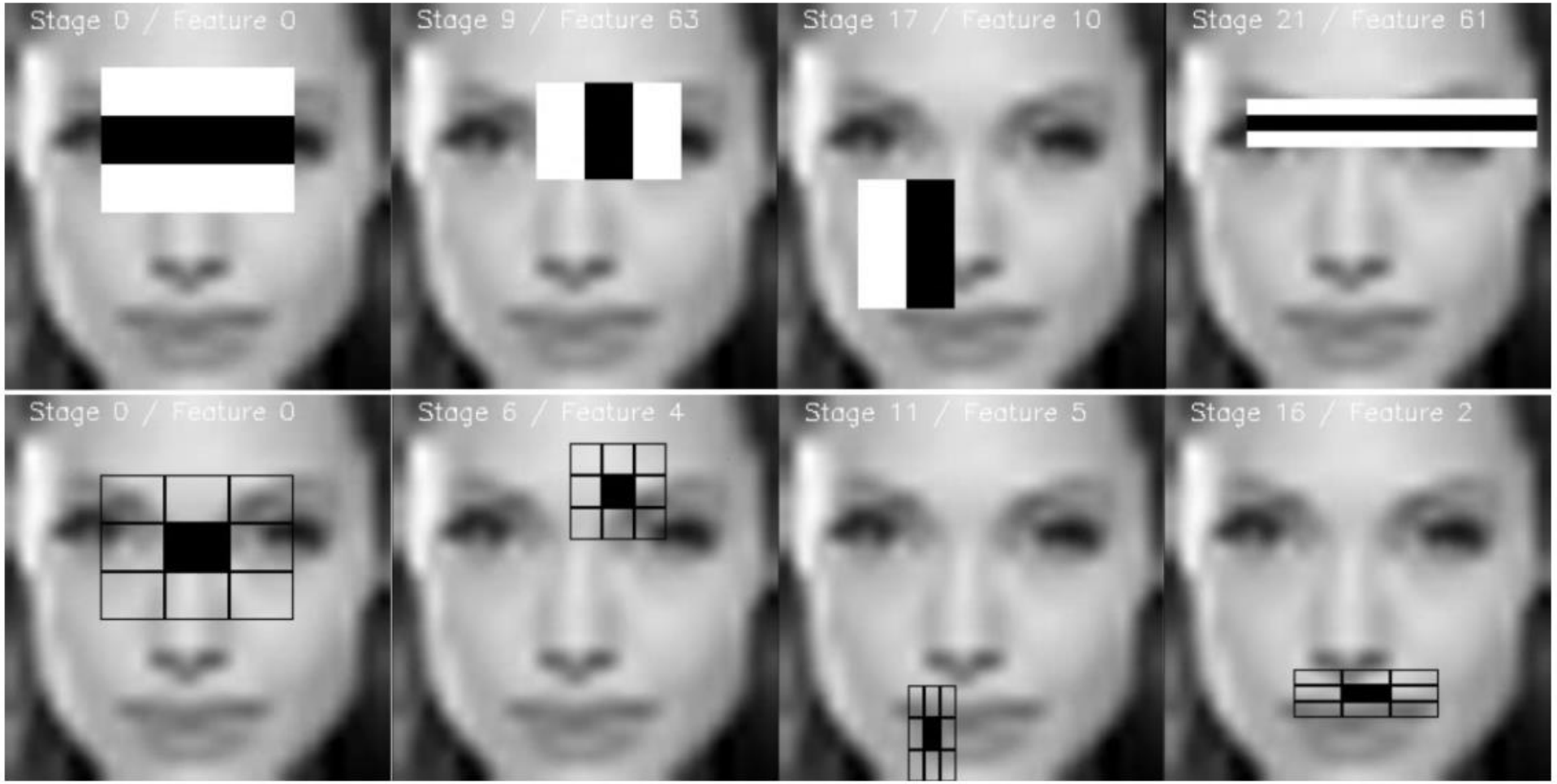


2000s: Emergence of Support Vector Machines and Viola-Jones Algorithm

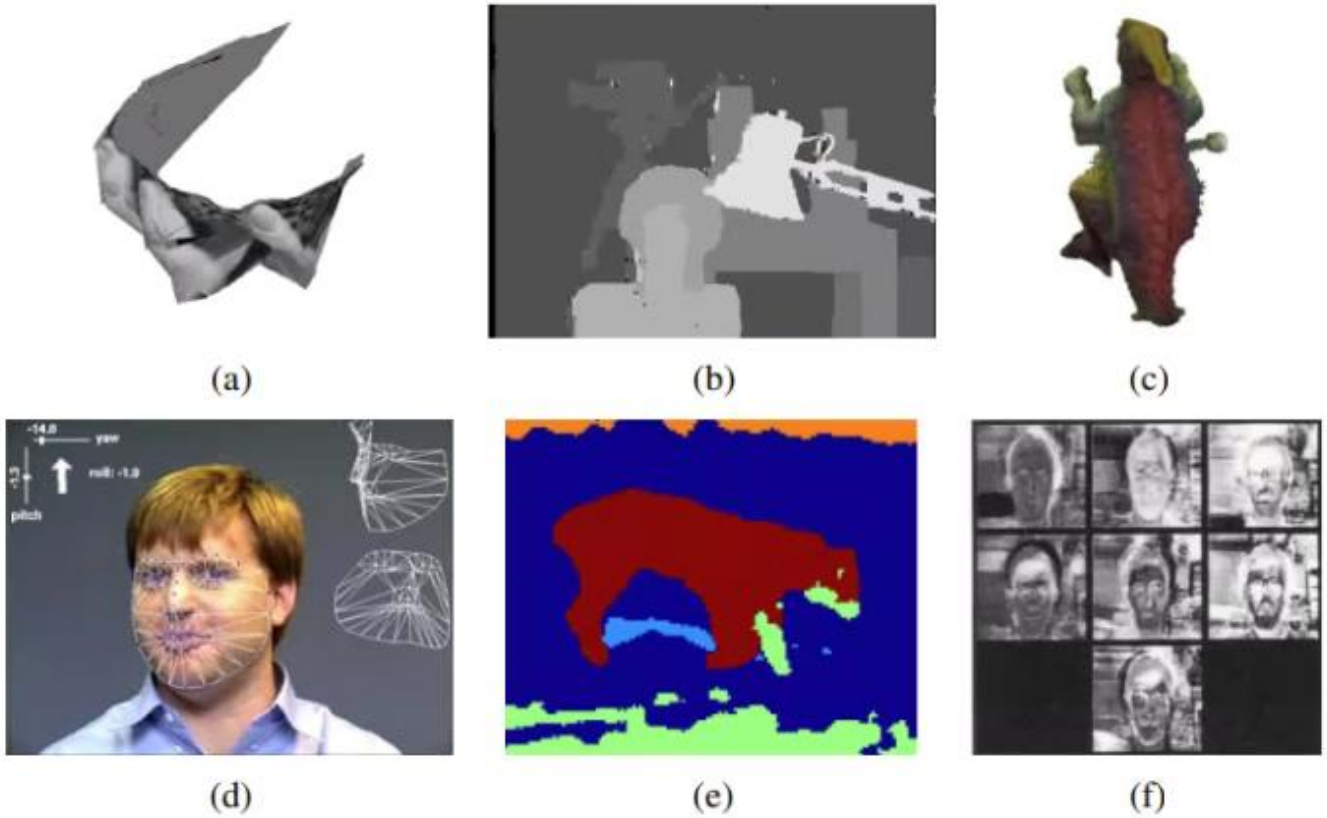




2000s: Emergence of Support Vector Machines and Viola-Jones Algorithm



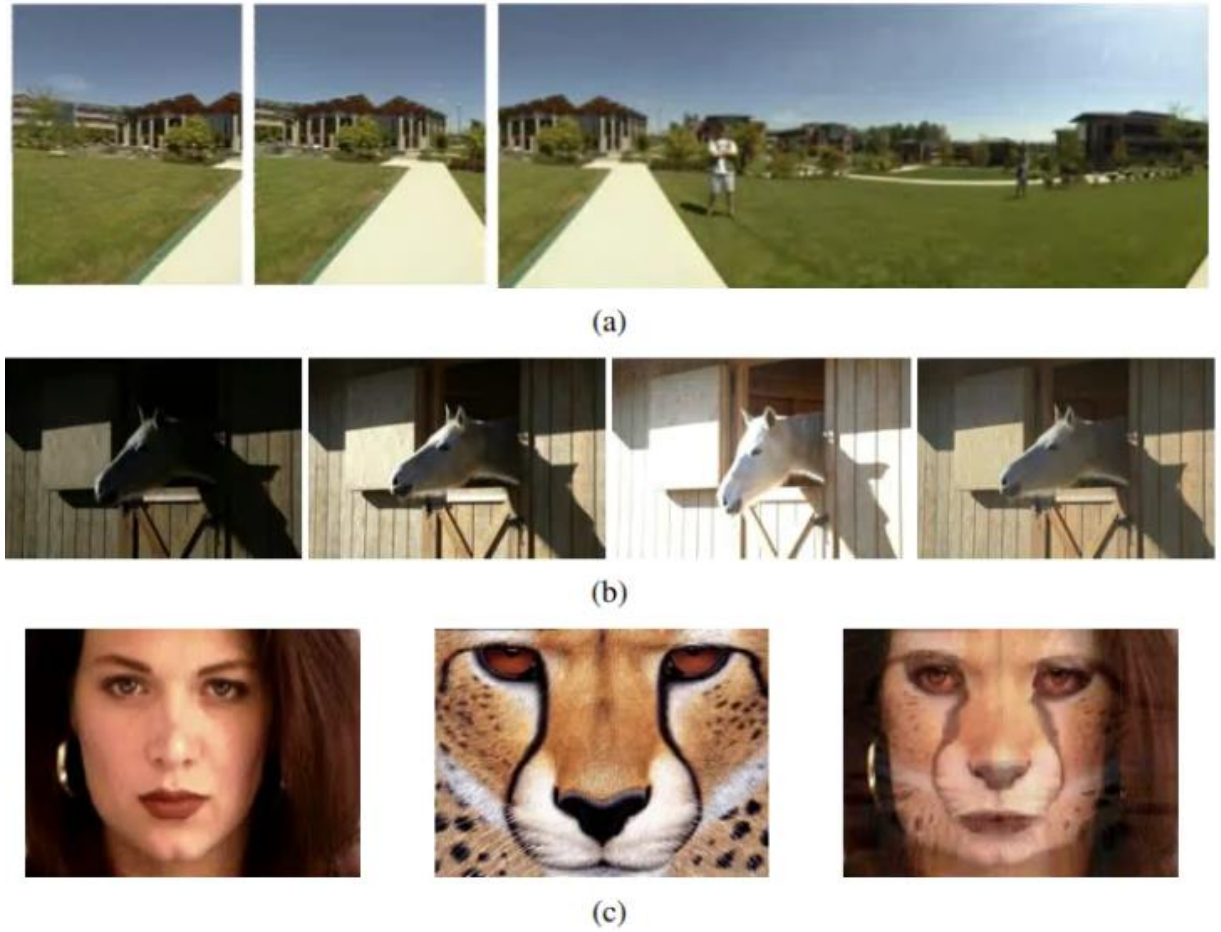
A brief history – 1990s-2000s



(a) Factorization-based structure from motion (Tomasi and kanade, 1992) ; (b) Dense stereo matching (Boykov, Veksler and Zabih, 2001) ; (c) Multi-view reconstruction (Seitz and Dyer, 1999) ; (d) Face tracking (Matthews, Xiao and Baker, 2007) ; (e) Image segmentation (Belongie et al., 2002) ; (f) Face reorganization (Turk and Pentland, 1991).

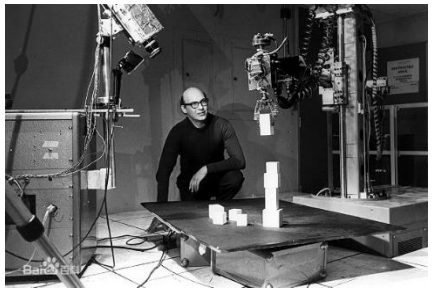
Szeliski R. Computer Vision: Algorithms and Applications[J]. 2011.

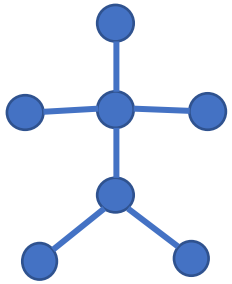
A brief history – 1990s-2000s

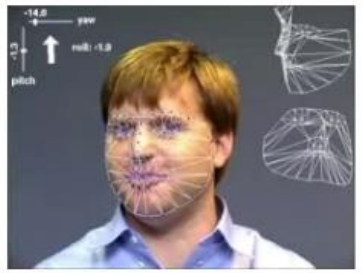


(a) Image stitching: merging different views (Szeliski and Shum 1997) ; (b) Exposure bracketing: merging different exposures; (c) Morphing: blending between two photographs (Gomes et al. 1999).

A brief history – 1960s-2000s







Observations on Small Samples



MINIST Handwritten Digits



Olivetti Face Dataset, AT&T

A brief history – 2010

ImageNet Large Scale Visual Recognition Challenge (ILSVRC)

1,000 object classes
1,431,167 images



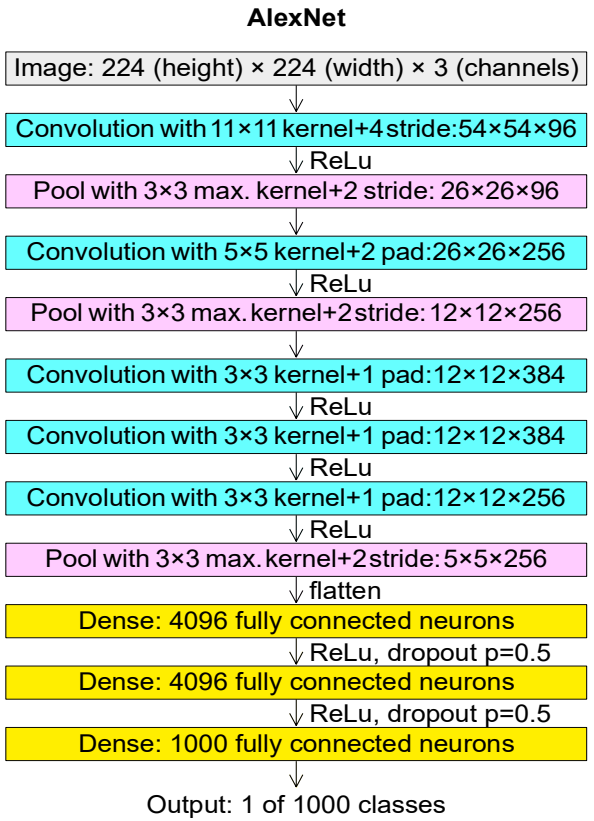
https://cs.stanford.edu/people/karpathy/cnnembed/

Krizhevsky A, Sutskever I, Hinton G E. ImageNet classification with deep convolutional neural networks[J]. Advances in neural information processing systems, 2012, 25: 1097-1105.

A brief history – 2013

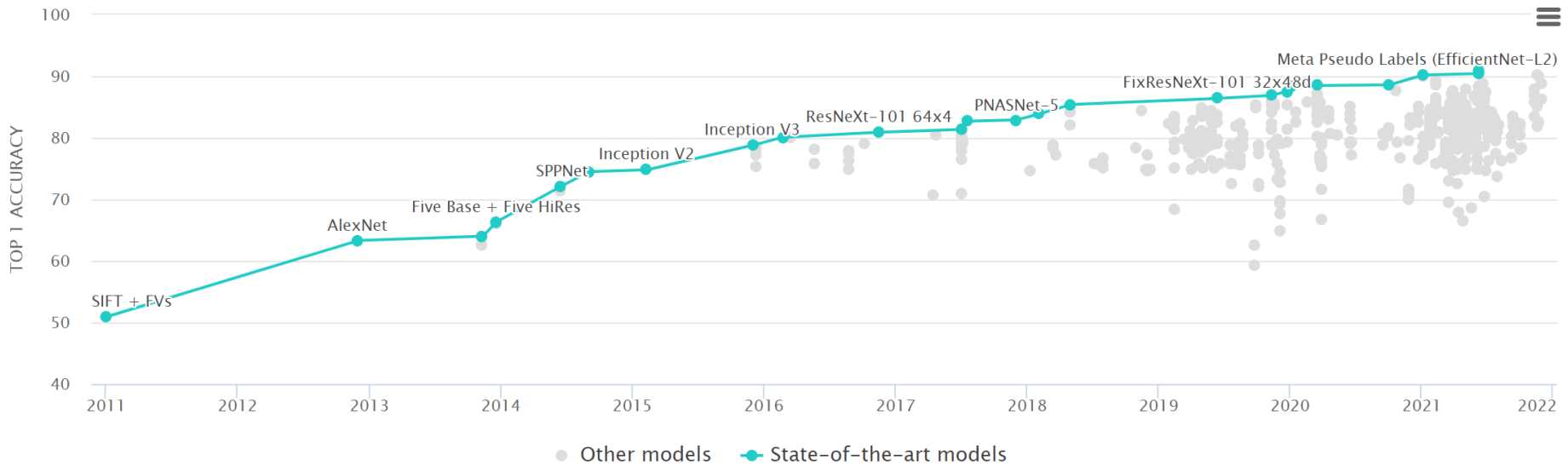
AlexNet
Winner in ILSVRC 2012

Top-5 error rate (rate of not finding the true label of a given image among its top 5 predictions) of 15.3%. The next best result trailed far behind (26.2%).



https://en.wikipedia.org/wiki/AlexNet
Krizhevsky A, Sutskever I, Hinton G E. Imagenet classification with deep convolutional neural networks[J]. Advances in neural information processing systems, 2012, 25: 1097-1105.

A brief history – 2010s-2020s





What is **Computer Vision**?

Keywords

- Recognition, Classification, Segmentation, Detection, Image Processing, Generation, Reconstruction, Semantics, Machine Learning, Models, Deep Learning …
- Large (Language) Models …



Learning CV by playing it



In-class Social Network Mining



Do you like attendance-check?



No!!!



Why don’t we make it fun?

An Interesting Way for Attendance-Check

- Take photos in class
- Publish them on course homepage
- Online check (by students) when downloading course materials

An Interesting Way for Attendance-Check



Photos taken from class

An Interesting Way for Attendance-Check



Image Stitching

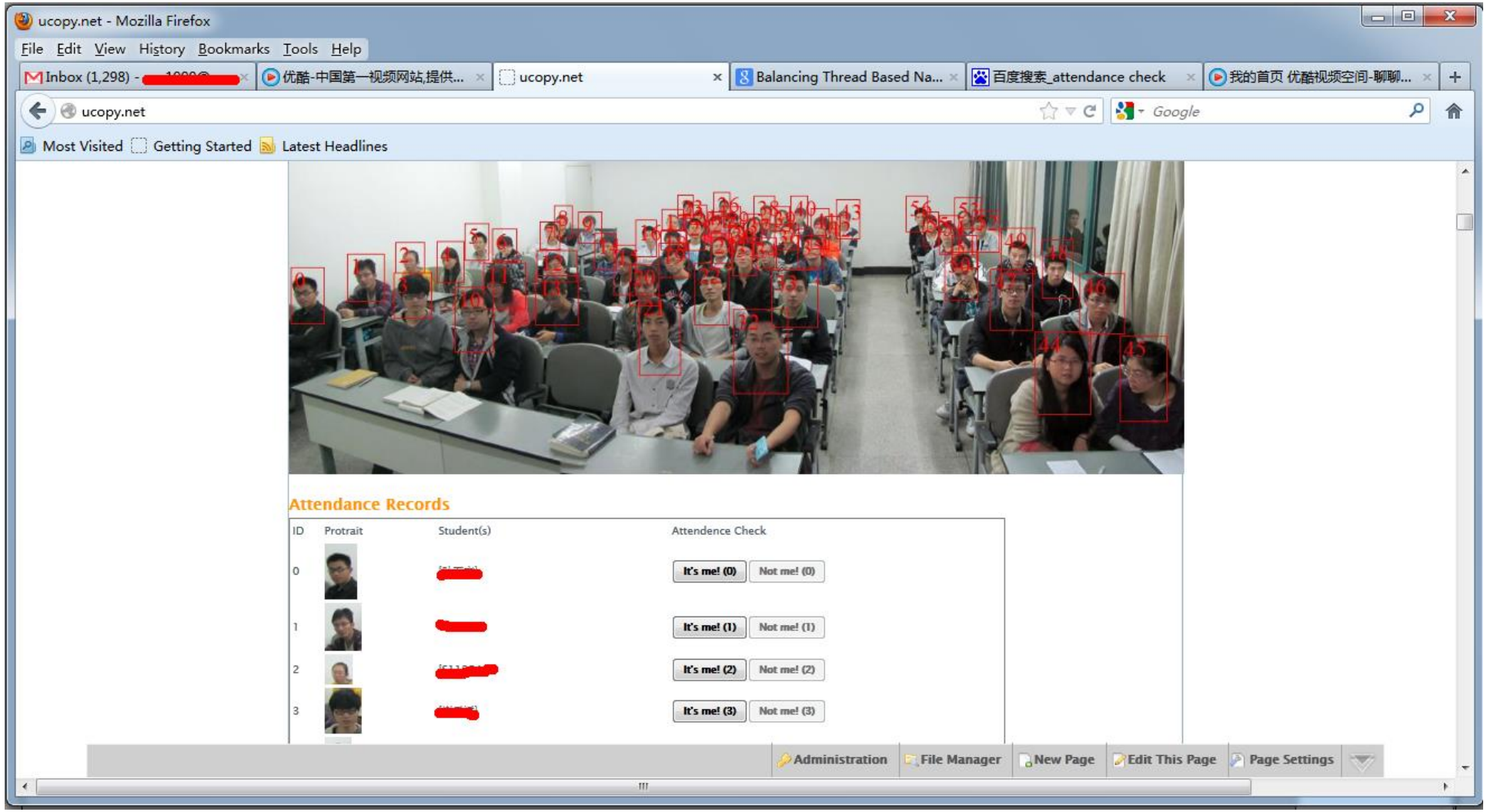


Figure 2: Challenges for face detection/recognition in an uncontrolled environment.

An Interesting Way for Attendance-Check



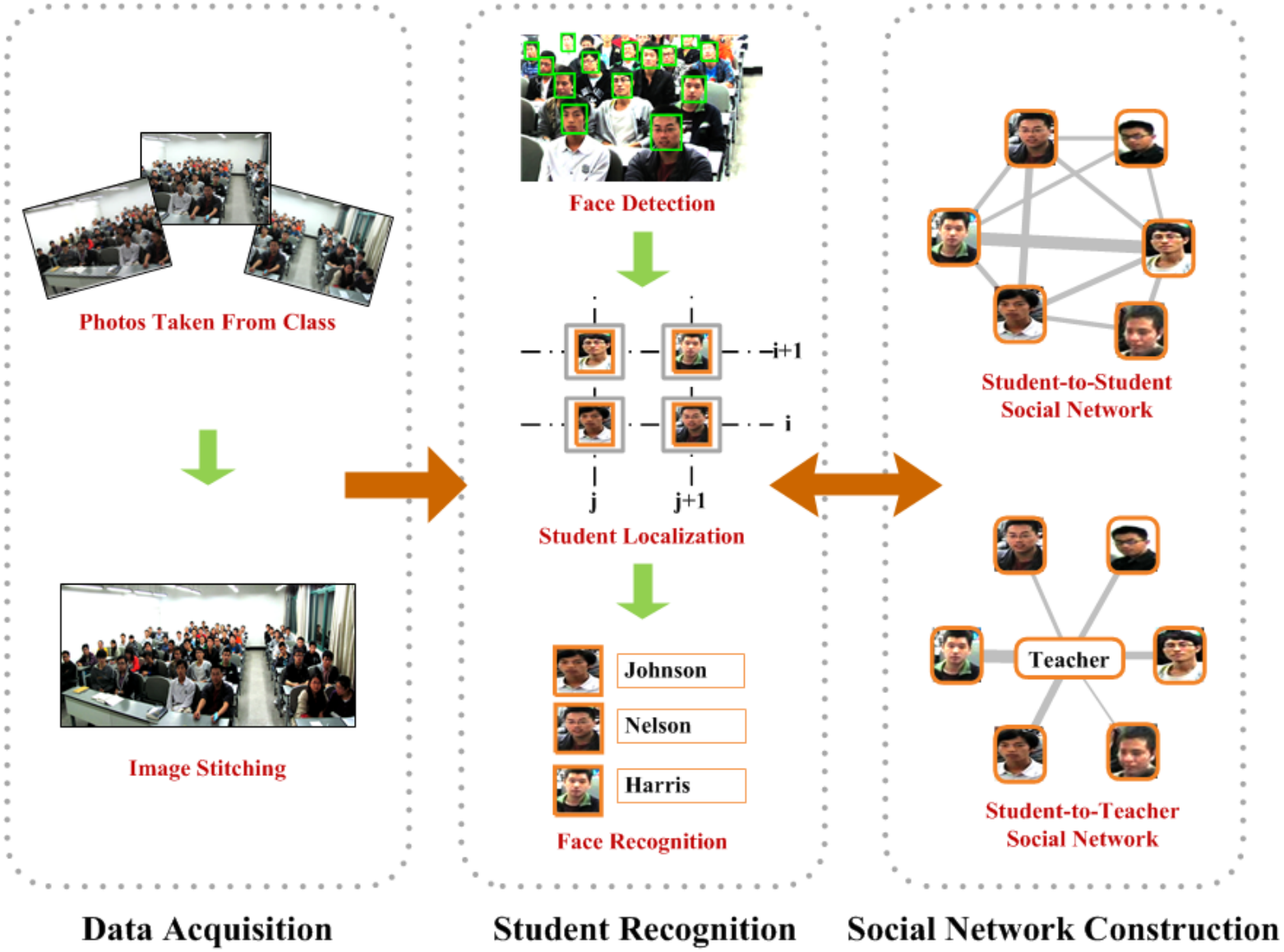
Face Detection



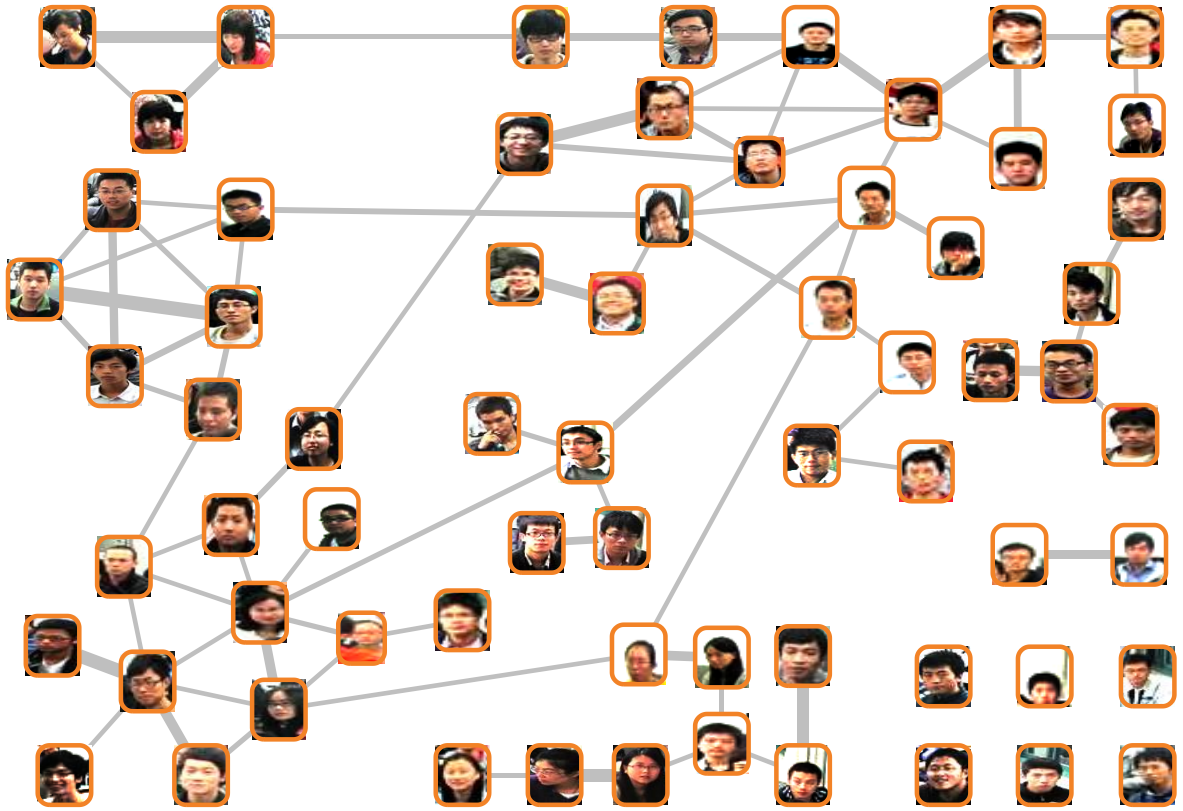
Online Check

Take a step further

- Integrate with facial recognition
- Conduct in-class social network analysis

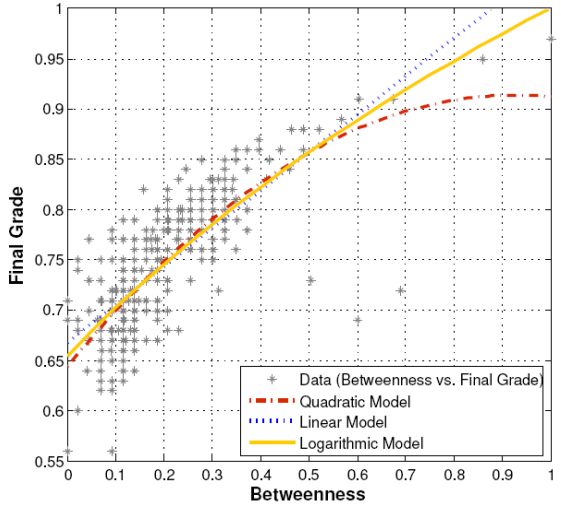

Photos Taken From Class
Image Stitching
Face Detection
i+1
i
j
j+1
Student Localization
Johnson
Nelson
Harris
Face Recognition
Student-to-Student
Social Network
Teacher
Student-to-Teacher
Social Network
Data Acquisition
Student Recognition
Social Network Construction

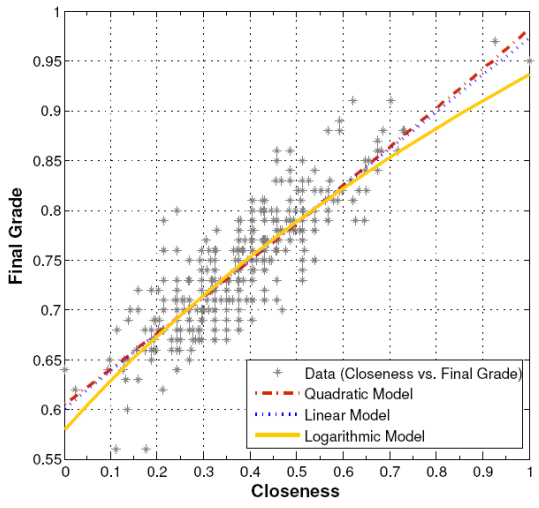
Social Network Construction & Pedagogical Analysis

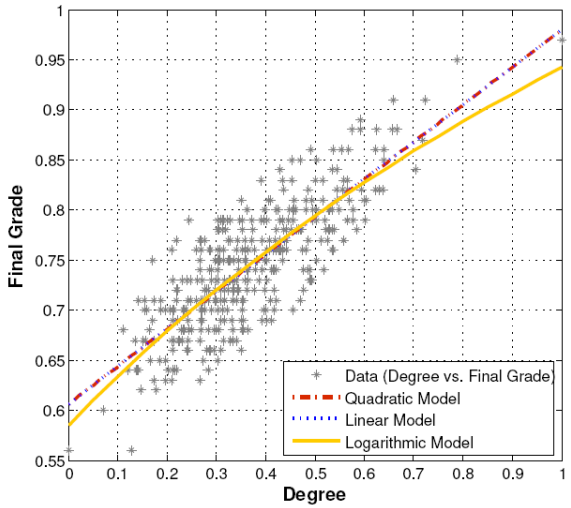


Social Network Construction & Pedagogical Analysis

- Student-to-Student network construction
  - Experimental Results

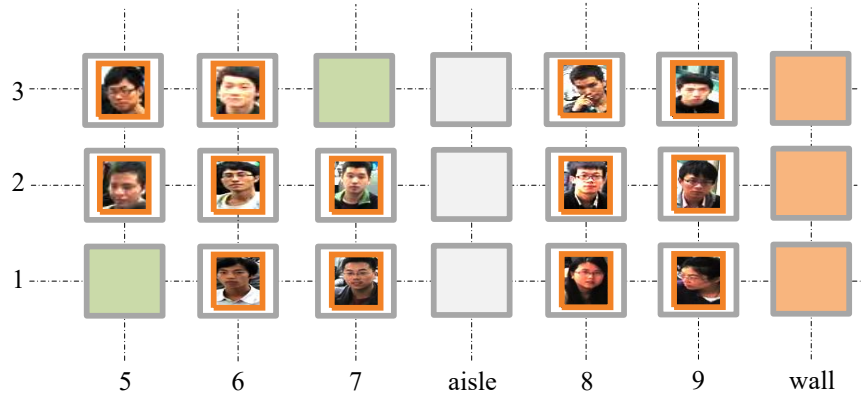




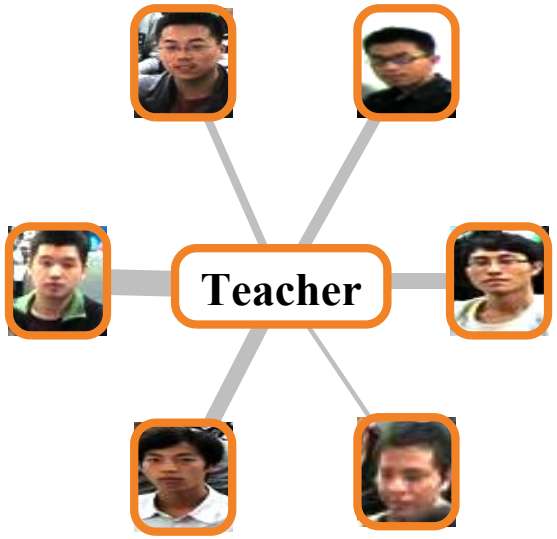


Social Network Construction & Pedagogical Analysis

- Student-to-Teacher network construction



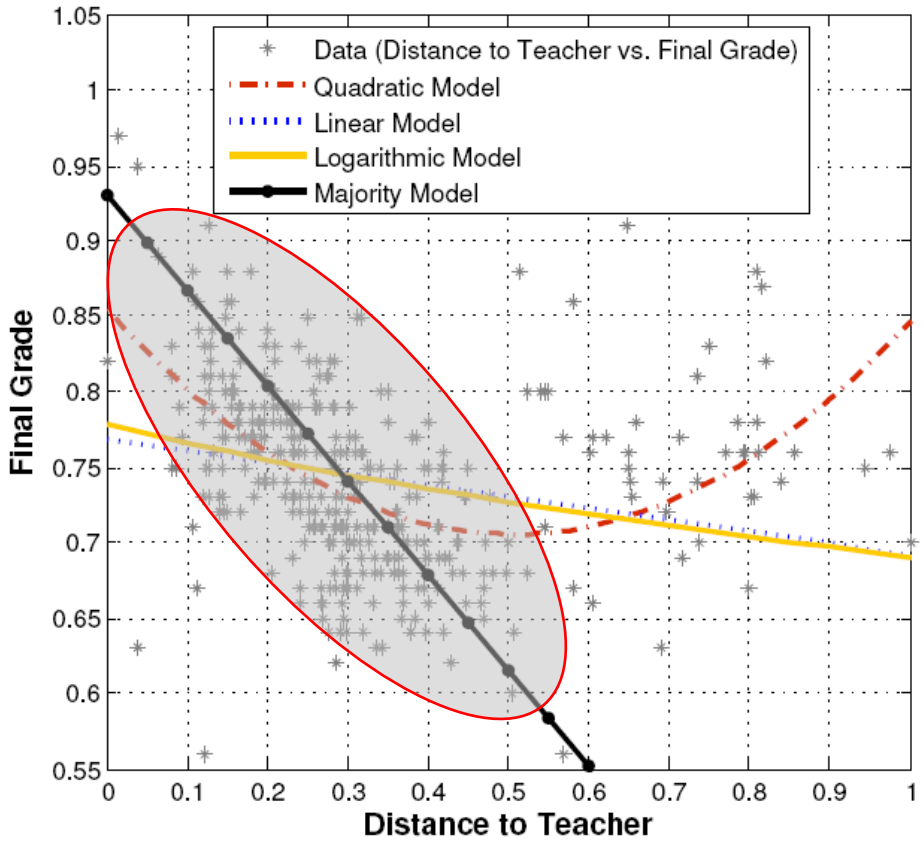
Teacher



Student-to-Teacher Social Network

Social Network Construction & Pedagogical Analysis

- Student-to-Teacher network construction



Conclusions

- Facial recognition in extreme conditions can be improved by social inference
- In-class social network is useful for pedagogical analysis to reach objective and quantitative conclusions

Mining in-class social networks for large-scale pedagogical analysis

XY Wei, ZQ Yang
Proceedings of the 20th ACM international conference on Multimedia, 639-648

Attendance-check can be fun !

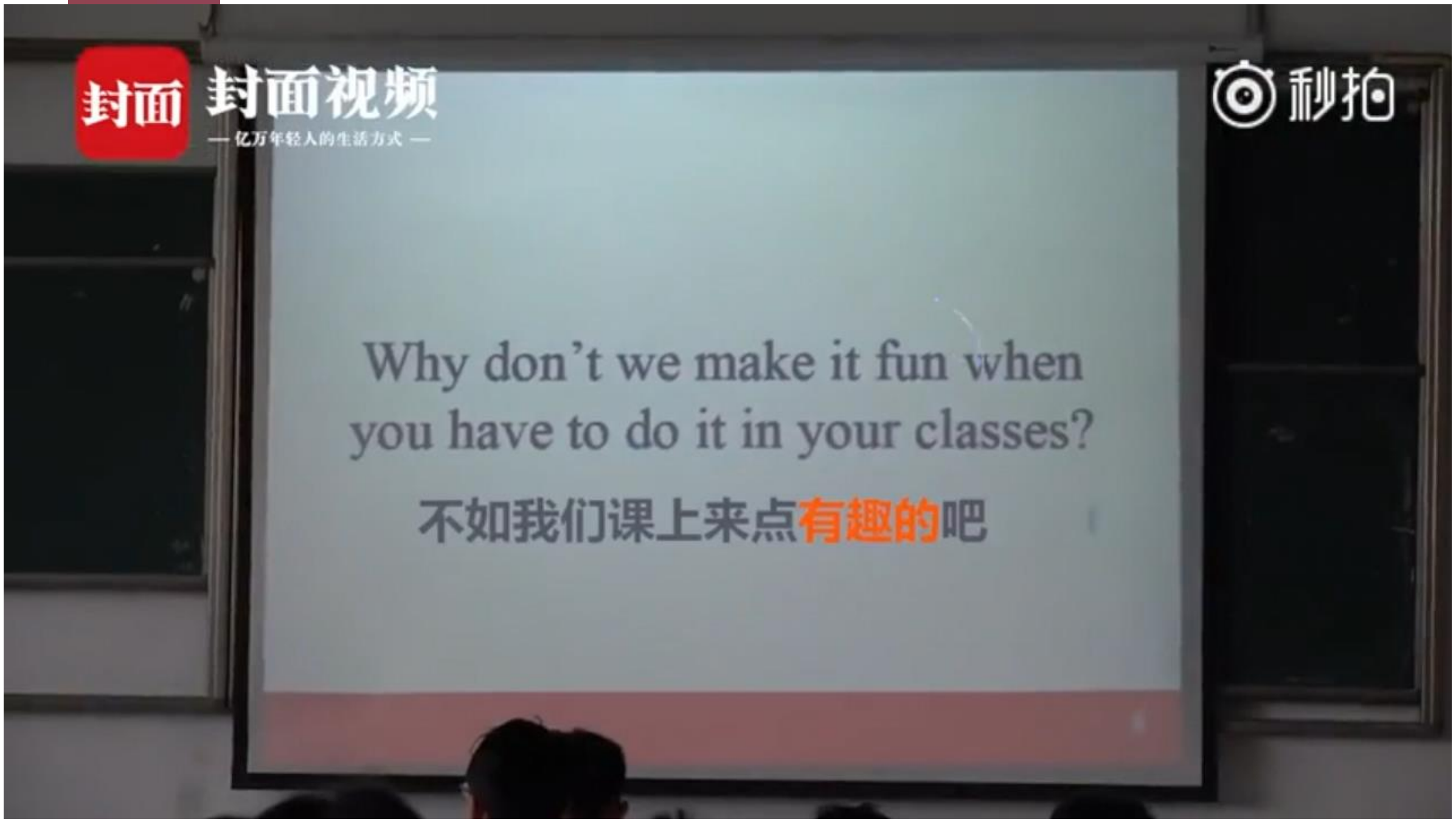






More steps further

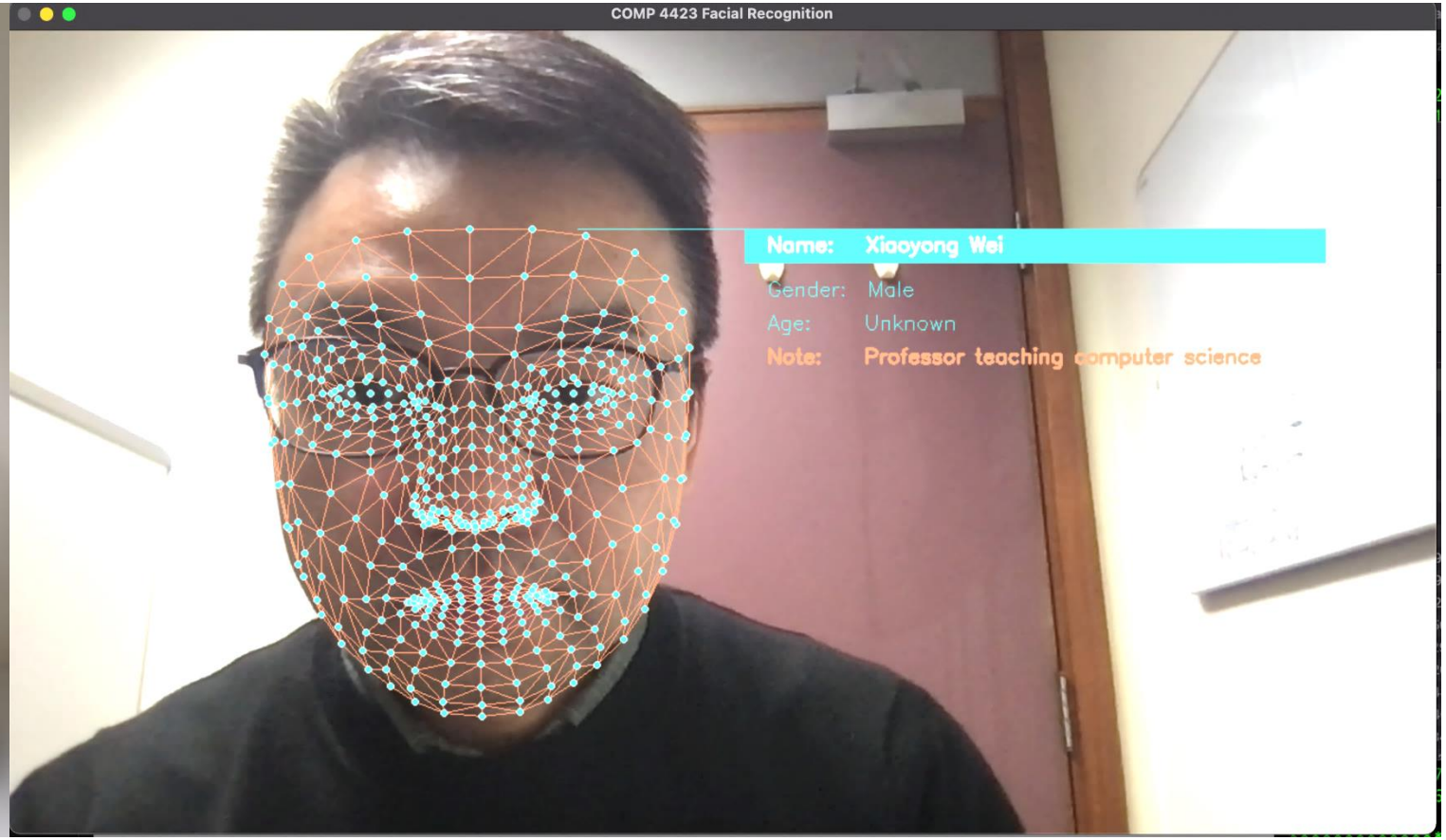

Multimedia Computing Lab (McLab)
Sichuan University (cswei@scu.edu.cn)
happy
neutral


封面
封面视频
— 亿万年轻人的生活方式 —
秒拍
Why don't we make it fun when
you have to do it in your classes?
不如我们课上来点有趣的吧

A step we can take together

FPV Recognition @ 4423





Virtual Campus Tour @ 4423

12:00



Introduction:

LibCafe:
A coffee shop in the YK Pao Library,
where you can buy good coffee to help you refresh your mind.

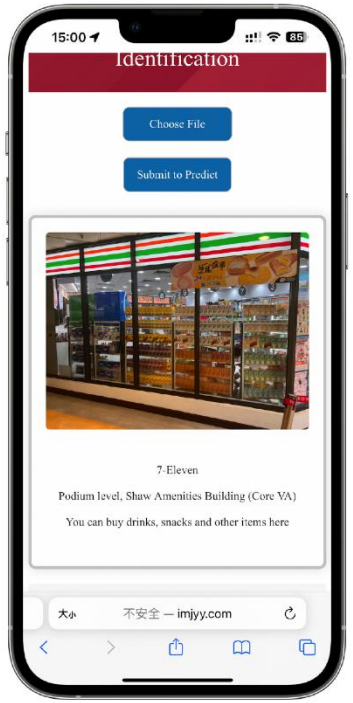
12:00

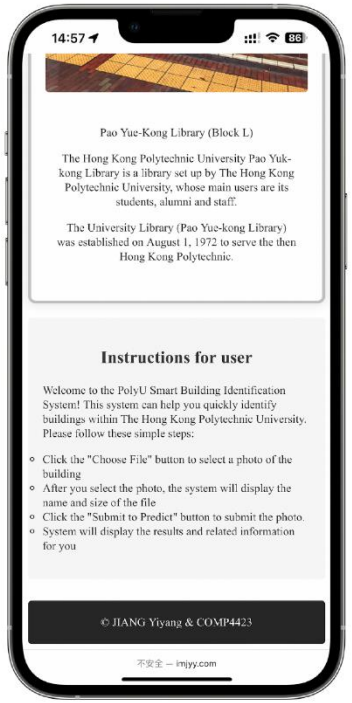


Introduction:

Jockey Club Innovation Tower:
It is designed by Pritzker-prize-winning architect Zaha Hadid.
It is now the headquarters of the School of Design.

Virtual Campus Tour @ 4423

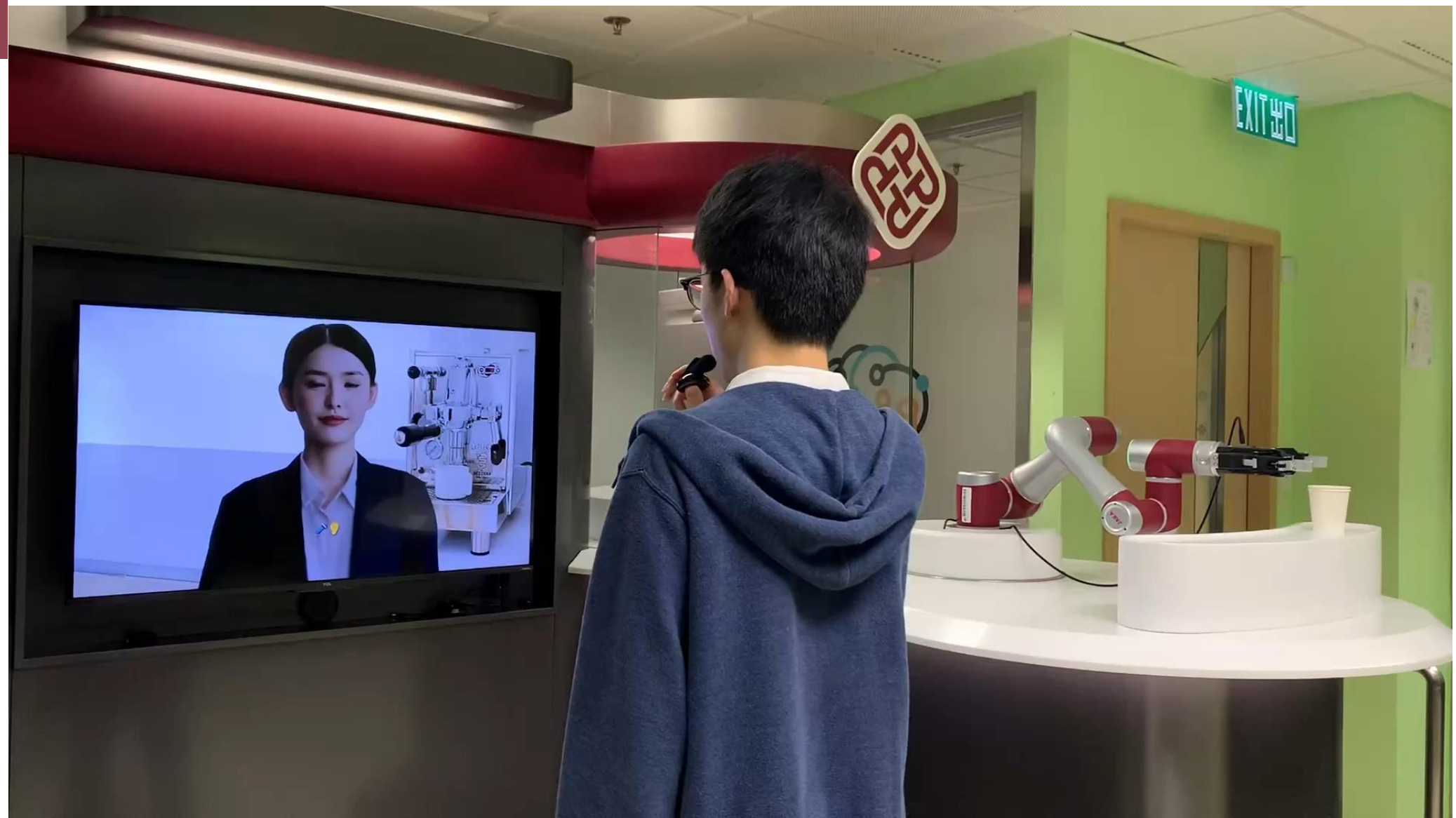




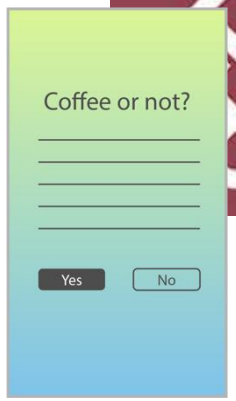



EXIT


EXIT


EXIT

THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學



Entrance

Guest selects on screen. If they want a cup of coffee or just browsing. Their waiting number will be display on screen.

Guest agrees to the T&C
Camera takes a photo of the guest's face who wants a coffee.

1



Call to the bar

Robot would call the guests to the bar. There are multiple spaces at the bar. The guest is guided to their spot with the screen on the bar that shows the guest's face.

2

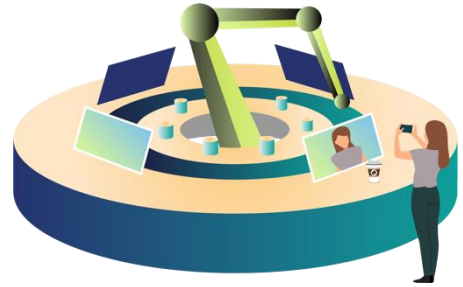
How about ...

Sure

3

Recommendation

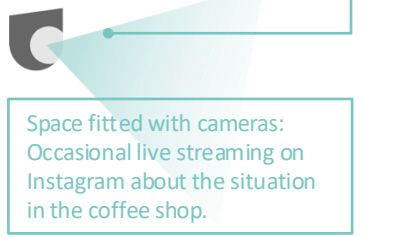
Chatbot chats with the guest then recommends a coffee. Guest accept or reject the suggestion and select their own coffee.



4

Serving

While the robot is making and serving the coffee. It plays small games/ group games with the guests. Coffee is served.



Space fitted with cameras:
Occasional live streaming on Instagram about the situation in the coffee shop.



6

Evaluation & data collection

Enjoy your coffee, evaluate the whole experience and coffee recommendation by filling a short questionnaire on tablet

5



Robot the coffee influencer

Occasional live streaming on Instagram and the big screen about the situation in the coffee shop.




NESCAFÉ
Alegria












Road Map (tentative)

- L01: Introduction
- L02: Image Processing 1
- L03: Image Processing 2
- L04: Feature Extraction
- L05: Image Retrieval
- L06: Image Classification 1
- L07: Image Classification 2



Road Map (tentative)

- L08: Advanced Image Retrieval
- L09: Advanced Image Classification
- L10: 3D Representation and Reconstruction
- L11: Advanced topics & Course projects
- L12: Advanced topics & Course projects
- L13: Advanced topics & Course projects
- Final Exam



Course websites

- PolyU Blackboard learn.polyu.edu.hk
  - Announcements
  - Slides
  - Assignments
  - Projects



Timetable

- Lectures at **QR403**
  - Tuesday. 9:30-11:20
  - Week 1-13
- Tutorials
  - **TUT001**: Tue. 11:30-12:20 at **PQ604A**
  - **TUT002**: Tue. 11:30-12:20 at **PQ604C**
  - **TUT003**: Tue. 11:30-12:20 at **PQ604B**
  - Week 2-13

Assessments

- Continuous assessments 30%
  - 2 Assignments 15% (5% each, late submissions acceptable up to 3 days with 10% deduction per day, 5% participation)
  - 1 Project 15%
- Final Examination 70 %

- Zero tolerance to cheating and plagiarisms

Dear Students,

We are very pleased to see the strong interest in COMP4423 Computer Vision. With around 80 students enrolled and over 30 on the waitlist, we are excited to welcome you. At the same time, we encourage everyone to make a careful and informed decision about whether this course is the right fit for you this semester.

This announcement has three goals:

- to explain why we are confident this is a useful and interesting course,
- to help you evaluate whether it matches your background and goals, and
- to suggest what to do if you are still unsure.

1. Why COMP4423 is a valuable course?

We are confident COMP4423 will be both practically useful and academically rewarding:

- **Sustained demand:** Since the course was first offered in 2022, the waitlist has been consistently long. This year, we increased the class capacity from 60 to 80 to accommodate more students.
- **Strong student feedback:** The course SFQ has remained high, typically in the 4.5-4.9 range.
- **Proven outcomes:** Past students have successfully applied what they learned in this course to achieve strong academic and research outcomes, including 1 Champion of 21st IEEE (HK) Computational Intelligence Chapter FYP & PG Competition, 2 Champions of COMP Best Project Award Competition in 2024 and 2025, 3 top-conference papers (MM 2024, EMNLP 2025, AAAI 2026) and 2 top-journal papers (TMI, BIB), and CV related real-world applications (https://polysmartgroup.github.io/).
- **Graduate progression:** 3 former students earned PhD scholarships, including 2 HKPFS awardees. In addition, 1 former student has been shortlisted by the Graduate School for this year's HKPFS.

2. How to assess whether this course is right for you?

A) Recommended preparation
COMP4423 is highly hands-on. You should be comfortable with programming, especially in Python, and have a genuine interest and motivation in AI and machine learning.

B) Who will benefit most
This course is mainly suitable for two groups of students:

- Those who aim to pursue research, with possible plans for PhD study in the future.
- Those who want a deep understanding of computer vision to support a strong FYP and improve readiness for AI-related internships or jobs.

C) Workload and expectations
You should be ready to accept a meaningful challenge. Please consider your timetable carefully before enrolling.

D) When this course may not be a good fit
This course may be less suitable if your main goal is simply to fulfill a credit requirement. Given the workload, COMP4423 may not be the most "cost-effective" option for that purpose.

3. If you are unsure: attend the first two lectures

If you are not yet certain whether this course matches your background and expectations, **we strongly encourage you to attend the first two lectures (Tue, 08:30-10:30, QR404)**. These sessions will provide a high-level introduction to computer vision, explain the course structure, and help you judge whether you are ready.

Thank you again for your interest in COMP4423. We hope this course will be a valuable experience for those who choose to take on the challenge, and we look forward to meeting you in the first lecture.

Best regards,
Teaching Team of COMP4423

https://polysmartgroup.github.io/

eSFQ 2021/2022

OPEN-ENDED COMMENTS ON THE SUBJECT

Note: Each bullet point denotes a response from an individual student.

1 What aspects of the subject were most useful to your learning?

- Practices of Computer Vision tasks and concepts on different Deep Learning mechanisms.
- Very interesting PowerPoint slides. Many advanced CV techniques introduced, many useful resources provided.

2 How could the subject be improved to help you learn better?

- NIL.

OPEN-ENDED COMMENTS ON THE STAFF MEMBER

Note: Each bullet point denotes a response from an individual student.

1 What aspects of this staff member's teaching were most helpful to your learning?

- Considerate about whether students caught up with the lecture content or not.
  Constantly change tutorial plans to adapt to student's need.
  Lecture contents created by him were great.
  Very nice teacher.
- Very patient and kind to students. Will explain difficult points over and over again until we can understand.

2 How would you like the teaching be changed (if at all), to help you learn better in the subject?

- NIL.

eSFQ 2022/2023

OPEN-ENDED COMMENTS ON THE SUBJECT

Note: Each bullet point denotes a response from an individual student.

1 What aspects of the subject were most useful to your learning?

- the hands-on experience for using different models and methods
- The flexibility of the professor and his willingness to receive feedback from the students as well as cater to both their needs and the needs of the industry. The tutorials and assignments were very helpful in applying what was learned in the theoretical classes.
- This subject is one of the most practical among my subjects. It not only contains basic theory but also self-learning and self hands-on practises, which enables student to transfer knowledge into real-world applications. I appreciate that we could have this kind of subject in UG study, and I wish we could more subjects like COMP4423.
- explained clear with images, give chance to try first and solution then given to explain
- Subject lecturer would go around during breaks and individually ask students if they understood everything. Lectures were relatively short (2 hours) so focus can be retained, but lecturer has shorter videos, 40 min, useful for revision
- The knowledge is practical and experimental, instead of just reciting. I like the way professor deliver the knowledge.

2 How could the subject be improved to help you learn better?

- not really related to learning, but not sure to what extent are we expected to deliver for the assignment, feels like each assignment's workload is an individual project
- NA
- can provide hands on example on advanced topics
- Timing of the assessments may be spaced better.
- The time duration for the assignment could be longer.

OPEN-ENDED COMMENTS ON THE STAFF MEMBER

Note: Each bullet point denotes a response from an individual student.

1 What aspects of this staff member's teaching were most helpful to your learning?

- the explanation is clear, other than some of the math stuff
- One of, if not the best instructors I have had at PolyU! From dynamically adjusting the course content to having a great understanding of the material in question and being able to explain/ teach it in an approachable manner. PolyU would be way better of if he was a full-time Prof. (rather then visiting)
- The professor was very good at engaging with the students, asking and answering their questions as well as providing office hours to consult him with anything related to the course. These extra office hours proved very helpful to me and helped me elucidate a lot of the questions I had with the course.
- provide helps when countering problem
- Prof. WEI, he is a very nice teacher, who provides very clear and interesting teaching in every lectures, and he pays lots of attention on students' feedbacks and voices, which makes a great balance between teaching and learning.
- I mentioned it before

2 How would you like the teaching be changed (if at all), to help you learn better in the subject?

- can probably provide more time for the assignments, or move them to before the end of the semester
- NA
- give more time and support to do the assignment
- NA

eSFQ 2023/2024

OPEN-ENDED COMMENTS ON THE SUBJECT

Note: Each bullet point denotes a response from an individual student.

1 What aspects of the subject were most useful to your learning?

- very few
- computer vision
- Very interesting teaching content.
- na
- good teahcer
- Pratical Experiences

2 How could the subject be improved to help you learn better?

- Make tutorial even more detailed; explain a systemetic way to learning coding for Computer Vision line by line so that we can even know how to code without seeing the reference code.
- good teacher
- more hands on tut
- Need more time to do assignments.
- na
- maybe more details

OPEN-ENDED COMMENTS ON THE STAFF MEMBER

Note: Each bullet point denotes a response from an individual student.

1 What aspects of this staff member's teaching were most helpful to your learning?

- indepth understanding in computer vision and teaching
- his ppt is fun
- na
- Very patient to answer questions.
- good teacher
- make the subject very interesting

2 How would you like the teaching be changed (if at all), to help you learn better in the subject?

- good teacher
- Need more time to finish two assignments.
- na
- no
- the assignment requiremnet should be more clear

eSFQ 2024/2025

OPEN-ENDED COMMENTS ON THE SUBJECT

Note: Each bullet point denotes a response from an individual student.

1 What aspects of the subject were most useful to your learning?

- all
- Good content and good present. Only thing is there could be more detailed explain provided.
  assignment is carefully designed, but actually is it not really associate to the lecture. However, a lot can be learned through the assignment and project.
  one of the best courses in this semester.
- The advanced cv technology and lectures and I will recommend everyone to take this course
- The comprehensive knowledge on different structures and models in the field of computer vision

2 How could the subject be improved to help you learn better?

- nonw
- No, everything is fine and well organized
- The tutorial is too hard and less explanation
- Cannot

OPEN-ENDED COMMENTS ON THE STAFF MEMBER

Note: Each bullet point denotes a response from an individual student.

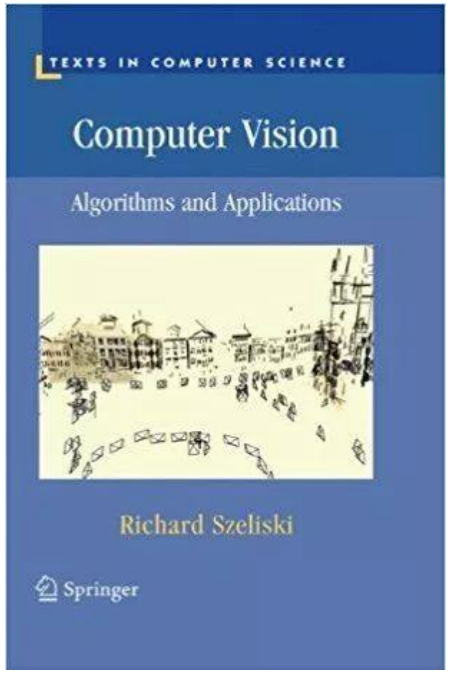
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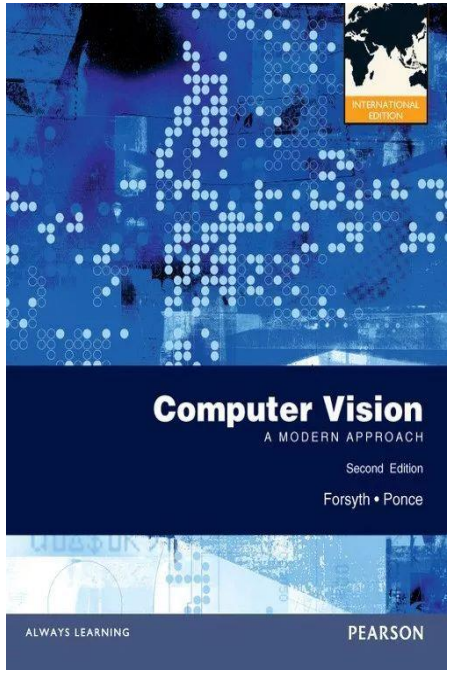
- His examples during lectures help me turned complex knowledge into easy ideas
- The insight into the AI technology
- all

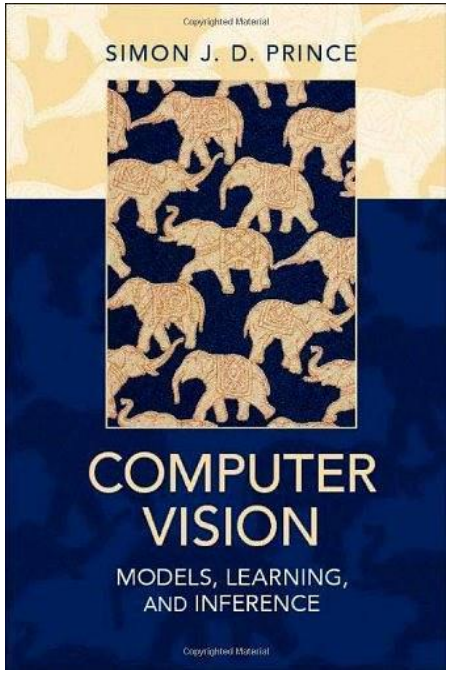
2 How would you like the teaching be changed (if at all), to help you learn better in the subject?

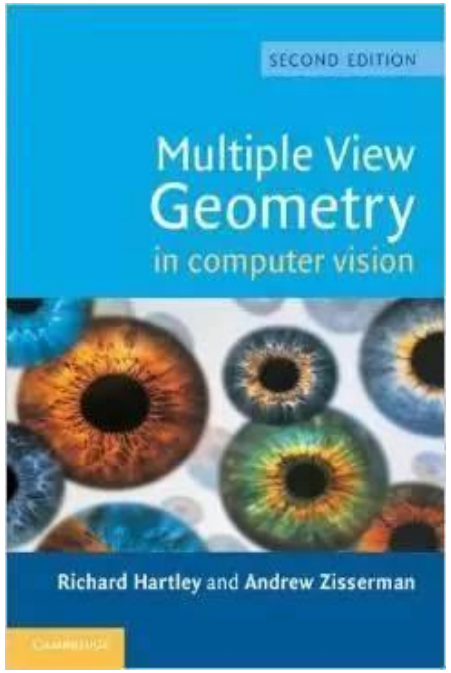
- nene
- Cannot
- No

References









Szeliski R. Computer Vision: Algorithms and Applications[J]. 2011.
Forsyth D, Ponce J. Computer vision: A modern approach[M]. Prentice hall, 2011.
Prince S J D. Computer vision: models, learning, and inference[M]. Cambridge University Press, 2012.
Andrew A M. Multiple view geometry in computer vision[J]. Kybernetes, 2001.



PolySmart Group

More at

https://polysmartgroup.github.io/

Thank You!

